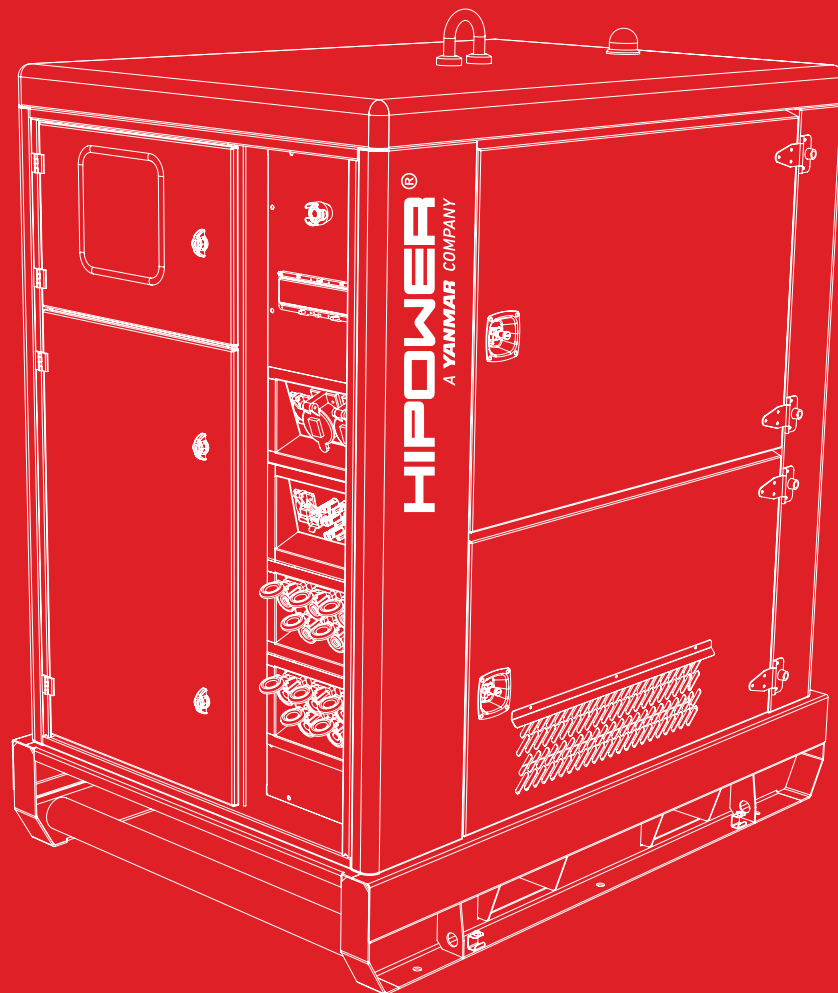




OPERATING MANUAL

EHR | Battery Power Generator

Single & Three Phase Models
Up to 40 kVA and 60 kVA energy storage

For Parts and Service Support Contact:

HIPOWER SYSTEMS

16600 S Theden St.
Olathe, KS 66062-9607

(866) 710-2988

Parts@hipowersystems.com
Service@hipowersystems.com



HIPOWER®
A YANMAR COMPANY

INDEX

3	1. INTRODUCTION
4	2. SAFETY STANDARDS
8	3. EHR OVERVIEW
15	4. UNLOADING, HANDLING AND TRANSPORT
19	5. EHR START-UP
43	6. MAINTENANCE
49	7. WARNING AND ALARMS
62	8. ANNEXES

1. INTRODUCTION

The aim of this manual is to provide you with information and basic instructions for the correct installation, transport, maintenance and use of your EHR.

It is essential that you carefully read all these safety rules and warnings before, during and after the start-up of your EHR: only in this way can we guarantee an optimal, regular service in perfect conditions of reliability and safety.

As this document is generic in nature, it is used for a wide range of products with a wide variety of options. Components, instructions or safety standards may appear that do not apply to or are insufficient for the specific equipment you have purchased. In that case you must assess and determine which instructions are valid for your equipment, supplementing them if necessary. Please contact Himoina's Technical Department if you have any doubts.

HIPOWER S.L. considers it honest to point out that the validity of the information described in this manual refers to the date of issuance of the same, since aspects such as technological progress and updates of current regulations, may force us to make modifications without prior notice.

This manual and the rest of the reference documentation are part of the EHR you have purchased and must be kept and protected against any agent that may deteriorate them. This documentation must accompany the equipment when it is transferred to a new user or owner.

You should always keep this manual close at hand, so you can consult it in case of doubt. Although the information provided in this manual has been verified in detail, HIMOINSA declines any responsibility derived from any calligraphic, typographical or transcription errors.

In accordance with the European Directives concerning the Defence of Consumers and Users, HIMOINSA is excluded from any liability as a result of incorrect installation, improper use of the machine or breach of the rules indicated in this manual.

2. SAFETY STANDARDS

Before working on the machine, it is important that you read the safety standards indicated in it carefully and that you be aware of any local requirements established in terms of safety.

Installation, operation, maintenance and repairs should only be carried out by authorised and competent personnel, and it is the responsibility of the owner of the EHR to carry out these operations in safe conditions. Parts and accessories should be replaced if they are not in safe operating condition.

As a premise to the content of this manual, the following are the basic criteria that, for your own safety and that of third parties, you must follow with special attention.

2.1 GENERAL SAFETY PRECAUTIONS

For your own safety and that of third parties, pay special attention to the following basic safety criteria:

- Do not allow any unauthorized persons or anyone using a pacemaker to access the EHR, due to possible electromagnetic interference in cardiovascular stimulator devices.
- Do not approach the EHR with loose clothing or objects that may be attracted by airflow or moving parts of the equipment.
- Do not smoke or cause any sparks in the vicinity of the EHR.
- It is forbidden both to cancel and/or disassemble the safety devices and to modify the configuration of the EHR.
- It is forbidden to lean on the EHR or place objects on it.

In the case of automatically operated equipment, it is also recommended that you:

- Place a red light in a clearly visible position and turn it on when the equipment is running.
- Place a danger sign warning about the possibility of the machine starting up unexpectedly.
- Place a mandatory sign stating that "All maintenance operations must be performed with the equipment turned off and the primary power supply disconnected."
- To perform an emergency shutdown of the equipment, press the

- “emergency stop” button, located on the outer panel next to the protective box.

2.2 ENVIRONMENTAL SAFETY

- Respect the rules and regulations relating to acoustic installations.
- Waste maintenance, storage and disposal operations must be carried out in accordance with the regulations of the country in which the equipment is being used.

2.2.1. FACTSHEET ON THE ENVIRONMENT AND WASTE DISPOSAL IN ACCORDANCE WITH EN 82079-1.

Waste disposal work related to the generator should only be carried out by suitably qualified personnel. Suitably qualified personnel are staff who, thanks to their training and experience in EHR maintenance and repair, know about the risks to health and the environment that may derive from the equipment and parts related to the equipment.

Before starting work, a qualified electrician should check electrical safety. Five safety standards must be respected:

1. Disconnect the installation. (Disconnect the voltage)
2. Secure against reconnection
3. Check that there is no voltage
4. Grounding and short circuit (Installations above 1000 volts)
5. Cover or separate live adjacent parts

The safety data sheets of all equipment must be consulted and respected.

Improper disposal may give rise to the following health and environmental risks:

- Burns
- Chemical injuries
- Poisoning
- Bruises
- Soil pollution
- Water pollution
- Air pollution

Ask your local authority responsible for waste disposal about the appropriate recycling points. To remove a generator, disassembly without dismantling is recommended.

2.3 SAFETY AND INFORMATION STICKERS

Distributed around the EHR are safety stickers and information designed to draw the attention of the operator or maintenance technician to potential hazards and explain how to act safely.

A brief explanation of the location is given below, together with information for each one.





Some of the stickers referred to here may not be needed for your EHR model and may not be included in the equipment.

2.4 WARNINGS IN THE EVENT OF MISUSE

The EHR supplied by HIMOINSA is intended for the conversion and optimisation of electrical energy according to the environmental and operational conditions and limits indicated or agreed in the contract. Any modification of these conditions and limits must be communicated to the factory directly or through authorised workshops designed to achieve optimal operation and, where appropriate, to provide new modifications and / or calibrations to the equipment.

The EHR is an energy conversion and optimisation system intended for energy distribution facilities that must be operated by specialists in accordance with current standards. Although the recharging powers may be lower than those of a public supply grid, the danger inherent in electric power is the same. The EHR is a production plant that, in addition to the electrical risks inherent in a power supply from a public supply grid or generator set, is also subject to the dangers derived from lithium batteries (explosion, gassing, spillage).

Any use that has not been previously agreed with HIMOINSA should be considered as improper use and, as such, is not acceptable.

2.5 HAZARDS AND WARNINGS

Loss of Life, Property damage, Installation must always comply with applicable codes, standards, laws and regulations. Failure to do so will result in death and serious injury.

GENERAL HAZARDS:

! DANGER

- Automatic starts-up. Disconnect utility power and render unit inoperable before working on unit. Failure to do so will result in death or serious injury.

! WARNING

- Risk of injury. Do not operate or service this machine if not fully alert. Fatigue can impair the ability to operate or service this equipment and could result in death or serious injury.
- Loss of Life. This product is not intended to be used in a critical life supports application. Failure to adhere to this warning could result in death or serious injury.
- Equipment damage. Only qualified service personnel may install, operate, and maintain this equipment. Failure to follow proper installation requirements could result in death, serious injury, and equipment or property damage.
- Personal injury and equipment damage. Do not use equipment as a step. Doing so could result in a fall, equipment damage, unsafe equipment operation, and could result in death or serious injury.
- Loss of control. Do not park the unit on a steep grade or slope, even if using a parking break. The unit could suddenly roll, move, or tip, which could cause death, serious injury, property or equipment damage.

Important Notes:

See individual unit specifications for required maintenance and run times pertaining to use.

- Connecting the EHR to the electric utility grid must only be done after receiving prior approval from the utility company.
- Protection against lightning surges in accordance with local electric codes are the responsibility of the installer.
- If working on this equipment while standing on metal or concrete, place insulative mats over a dry wooden platform. Work on this equipment only while standing on such insulative mats.
- Never work on this equipment while physically or mentally fatigued.

CAUTION

- Equipment or property damage. Do not block air intake or restrict proper air flow. Doing so could result in unsafe operation or damage to unit.

BATTERY HAZARDS:

DANGER

- Electrocution. Do not wear jewelry while working on this equipment. Doing so will result in death or serious injury.

WARNING

- Explosion. Do not dispose of batteries in a fire. Batteries are explosive. Electrolyte solution can cause burns and blindness. If electrolyte contacts your skin or eyes, flush with water and seek immediate medical attention.
- Electrical Shock. Disconnect battery ground terminal before working on battery or battery wires. Failure to do so could lead to death or serious injury.

FIRE AND EXPLOSION HAZARDS:

DANGER

- Fire and explosion. Installation must comply with all local, state, and national electrical building codes. Noncompliance could result in unsafe operation, equipment damage, death or serious injury.

ELECTRICAL HAZARDS:

DANGER

- Electrocution. Verify electrical system is properly grounded before applying power. Failure to do so will result in death or serious injury.

WARNING

- Personal injury and equipment damage. User access prohibited. Do not open the enclosure. No user serviceable parts inside. Only qualified service personnel may install, operate, and maintain this equipment. Noncompliance could result in personal injury or equipment damage.
- Electrical shock. Only a trained and licensed electrician should perform wiring and connections to unit. Failure to follow proper installation requirements could result in death, serious injury, and equipment or property damage.

3. EHR OVERVIEW

The EHR is a device that makes it possible to manage energy in stand-alone systems of gensets and / or renewable energy sources. In addition, when integrated with diesel generator systems, it optimises efficiency, noise, emissions and fuel consumption. It should be noted that it is a power generation system designed for the rental market.

The HICORE energy control module is the brain of the unit, consisting of a control panel that allows high-level unit supervision, monitoring and control. This control module, together with the C4CLOUD device that incorporates GPS global positioning technology, makes it possible to record and send data to the EHR PORTAL.

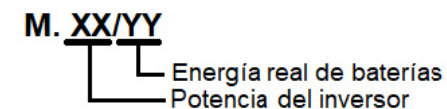
The EHR PORTAL is an energy management platform (EMS) that allows the user to control and optimise the energy consumed, in addition to monitoring and reporting on each equipment or the entire fleet of equipment owned by the user. This management can be done by both proprietary users and fleet managers and service engineers.

HICORE provides the best resources to manage your energy storage, allowing you to get the most out of the system and respond to your customers' needs more effectively.

This manual will guide you in the steps required to operate this equipment correctly and in a safe and efficient way, allowing you to manage the energy savings that the equipment will provide throughout its operational life.

3.1 OVERVIEW

The EHR equipment is available in several models and different versions are available for each of them. The models depend on the apparent power incorporated by the inverters and as a consequence of the energy available in the batteries. The versions are differentiated by the possibilities included in each model, in order to adapt to the demands of users.



The available models are as follows:

Model	Subject
EHR 20	Single-phase
EHR 30	Three-phase
EHR 40	Single-phase

Each EHR model contains 1 or 2 versions, and these vary depending on the maximum power they can withstand. Below is a table showing the “SOCKETS” each model come equipped with.

EHR 20 & 40

SOCKETS OPTIONS		240/120V/60Hz/2p+N+E
		V1
IN	120V 20A Waterproof Receptacle (HBL61CM64)	1
	400A 4/0 Single Pin Male Camlocks	L1+ L2+N+E
OUT	120V 20A Waterproof Receptacle (HBL61CM65)	1
	240V 50A Waterproof Twist Lock Connector (CS6369)	2
	400A 4/0 Single Pin Female Camlocks	L1+ L2+N+E

Every socket include circuit breaker. Power Output protected by Earth Leakage Relay. Domestic sockets with ELR

EHR 30

SOCKETS OPTIONS		208/120V 60HZ/ 3p + N + E
		V1
IN	120V 20A Waterproof Receptacle (HBL61CM64)	1
	400A 4/0 Single Pin Male Camlocks	L1 + L2 + L3 + N + E
OUT	120V 20A Waterproof Receptacle (HBL61CM65)	1
	240V 50A Waterproof Twist Lock Connector (CS6369)	2
	400A 4/0 Single Pin Female Camlocks	L1 + L2 + L3 + N + E

Every socket include circuit breaker. Power Output protected by Earth Leakage Relay. Domestic sockets with ELR

EHR equipment incorporates a control panel where the user will make all the connections and configure the equipment. This control panel is divided into 4 different parts:

A. Central

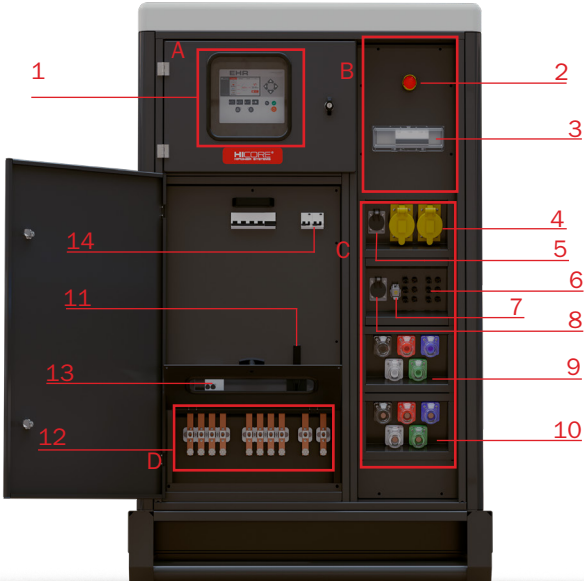
B. External protections

C. Input and output connectors

D. Panel of busbars

Each EHR element is listed below.

3.2 IDENTIFICATION OF THE MAIN COMPONENTS



Section	Number	Element
A. Control unit	1	HICORE display
	2	Emergency Stop button
B. External protections	3	Circuit Breakers: x2 20A 1 pole for 125VAC receptacles x2 50A 2 poles for 240/ 120VAC twist locks
C. Inlet and outlet connectors	4	x2 240/120VAC 50A 2 poles twist locks (Outlet)
	5	x1 120VAC 20A 1 pole receptacle (Outlet)
	6	Solar MC4 Connectors
	7	AutoStart Connection (Genset)
	8	x1 120VAC 20A 1 pole receptacle (Maintenance Charge) (Inlet)
	9	Power Output
	10	Power Input
	11	Input selector between busbar or "SOCKET"
D. Busbar panel	12	Input, output and busbar grounding
Protections	13	General output thermal motorised protection
	14	Input thermal and differential protection

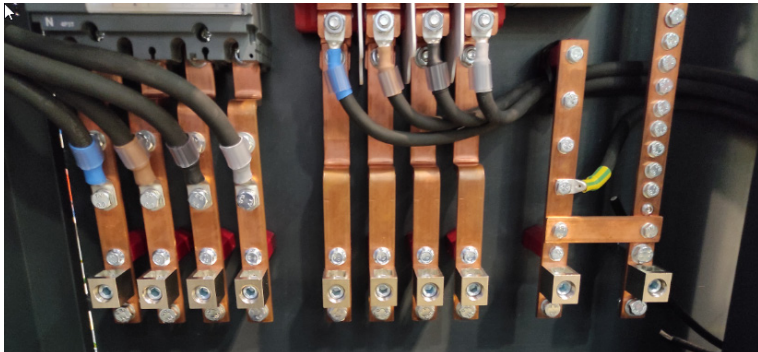
As you can see in the “Input and output connectors” section, EHR model versions may vary depending on the version.

There are several ways to connect external elements to the EHR to adapt it to the user’s needs. For this purpose, it incorporates a busbar panel and it allows the direct connection of the primary energy source (input), the installation to be supplied (output) and the grounding system. At the same time, it incorporates “SOCKET” input and output connectors for a faster connection.

This busbar panel is located behind the lower distribution door. It can be accessed by opening the lock located in the middle of the door.

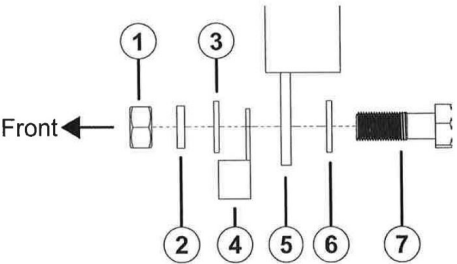


Once the busbar door is open, the user will only be able to access the connection terminals of each busbar. Attached below is the layout of each of the busbars.



The grouping of each of the busbars from left to right is indicated below.

- AC terminal output– (N ,L1, L2, L3): To attach 10 mm ring terminals.
- AC terminal input – (N ,L1, L2, L3): To attach 10 mm ring terminals.
- Ground node - Ground connection: To attach 10 mm ring terminals.



No.	Element
1	M10 Smooth nut
2	M10 Spring EHR
3	M10 Flat EHR
4	M10 Ring Terminal
5	Bar
6	M10 Flat EHR
7	Screw M10

It is mandatory to create a connection between the EHR and the ground, always in compliance with local rules and regulations. Depending on the grounding system to be implemented, a connection must be made to the ground or to the EHR. A chassis plate and a grounding plate are included and, depending on the grounding system to be wired, these plates must be bridged or not.

For a proper connection, one or more grounding rods must be wired. It should be noted that when a “grounding rod” is mentioned, this refers to a physical grounding connection. It is usually done by placing a copper bar (grounding rod) in the ground. But depending on the location of the system, it can also be a copper plate or strip buried in the ground, the water supply grid, or the water pipes of a house.

EHR equipment allows connection with TNS, TT and the IT grounding system. Listed below are the steps to follow to connect each of the grounding systems.

Identify the EHR by the appropriate label with its basic reference values, as well as the serial number, the unique identification for each unit:



3.3 TECHNICAL CHARACTERISTICS AND CAPACITIES

The EHR has some features depending on the model. Below is a table in which you can see the technical characteristics of each of the EHR models.

	EHR 20	EHR 30	EHR 40
PRIMARY POWER SUPPLY INPUT (THREE-PHASE)			
Current per phase of maximum power (A)	100	100	100
Supported input voltage range (WL)	[180 - 280]	[155 - 240]	[180 - 280]
Input frequency (Hz)	[55 - 65]		
Power. Maximum input (W)	38000	28700	76500
Generator set with automatic start-up	Yes		
MAINTENANCE INPUT (SINGLE-PHASE)			
Maximum maintenance power current (A)	20		
Input voltage	120 ± 2%		
Maximum maintenance power supply (W)	2400		
RENEWABLES INPUT (SOLAR)			
Maximum Maintenance Power Current (A)	8000	4000	8000
Maximum photovoltaic voltage (V)	240		
Photovoltaic short circuit current per MC4 connector (A)	23		
Number of MC4 solar connectors	3	3	6
OUTPUT (THREE-PHASE)			
Output Voltage Range (LYV)	240 ± 2%	208 ± 2%	240 ± 2%
Output frequency (Hz)	50 ± 0,1%		
Maximum phase current at output (A)	152	204	250
Three-phase power. Maximum output (W)	58000	96000	58500
Three-phase power of continuous inversion at output at 77°F (W)	20000	24000	40000
Three-phase power of continuous inversion at output at 104°F (W)	18000	21600	36000
Three-phase power of continuous inversion at output at 149°F (W)	13800	16560	27600

Maximum power Three-phase inversion, 0.5 s (W)	40000	60000	80000
Maximum inversion efficiency (%)	96		
Minimum power factor allowed at output	0.7		
BATTERIES (STORAGE)			
Battery technology	LFP (lithium iron phosphate)		
Maximum Usable capacity at 77°F (kWh)	42.6	56.8	56.8
Charging time at 77°F (h)	2.34	2.45	2.17
Maximum efficiency at 25°C (%)	90		
BMS Protections	Yes		
Life cycles (at 70% of their original capacity)	10000		
Maintenance load cycles	>1 Month		
GENERAL			
Three-phase power. Maximum battery charge (kW)			
Vacuum consumption without ventilation (W)			
Fan noise (dB)			
General working temperature (°C)	-4 - 140		

UPS function

The EHR can be used as an offline uninterruptible power supply (UPS). In the event of a grid or power supply failure, the EHR will continue to supply power to the output loads.

The change of status is less than 20 ms with zero interruption of the output supply, which means there will be no interruptions in power.

HICORE incorporates an operating mode intended exclusively for this function (see 5.10 Working modes | UPS mode). It requires no user intervention and will continue to power the output until the EHR power storage runs out. If the primary power supply is detected once again, it will supply power to charge the EHR battery to ensure it is ready for the next power outage.

Power assist

The Battery Assist (Power Assist) function is automatically activated when necessary, without the user having to take any action.

The power assist function is activated when the load demand rises above the input limit of the AC source. When activated, the inverters synchronize with the AC source and add the power to the incoming power. This feature is useful when a load demand is needed at intervals that the AC source cannot supply.

Each EHR model has a minimum power value that the primary power supply must have for the power assist to work. If it doesn't, the system will not run properly. Therefore, when choosing the primary power supply, note that these vary depending on whether it is an electricity distribution grid or a generator set:

• MINIMUM POWER ELECTRICAL GRID

	EHR 20	EHR 30	EHR 45
Minimum three-phase current (A)	4.00	11.00	15.00
Minimum power (kW)	3.00	8.00	10.00

• MINIMUM POWER GENERATOR SET

	EHR 20	EHR 30	EHR 45
Minimum power GENERATOR SET (kW)	20kW	30kW	45kW

ATTENTION:

The Power Assist (PEAK SAVING) function in combination with a generator set is limited to these generator set powers and in order to ensure correct use and avoid oscillations in the genset, it is recommended to perform it on 90% of the capacity of the generator set.

If the genset power is lower than recommended, HICORE will display an on-screen warning indicating that the generator set capacity is less than the minimum value.

3.4 DIMENSIONS AND BASIC DATA MODEL

MODELS 60HZ		EHR 20/20		EHR 20/40	EHR 30/30	EHR 30/60	EHR 40/60
Nominal power	kVA	20			30		40
Storage capacity	kWh	21.3		42.6	28.4	56.8	56.8
Voltage (60Hz)	VAC	240V/120V			208V		240V/120V
Lifespan (90% DoD)	Cycles	6000			6000		6000
Battery management system				LFP batteries with built-in BMS			
Recharge time / Maintenance time ⁽¹⁾							
Recharge time	h	1.17		2.34	1.28	2.45	2.17
Maintenance recharge (@DoD%)	h	5.59		11.17	11.17	14.90	9.77
Battery discharge time ⁽²⁾							
100% of te rated power	h	0.74		1.47	0.66	1.31	1.09
75% of te rated power	h	0.98		1.97	0.87	1.75	1.46
50% of te rated power	h	1.47		2.95	1.31	3.12	2.18
25% of te rated power	h	2.95		5.90	2.62	5.24	4.37
Dimensions (L x W x H)	in	65 x 43.30 x 71			65 x 43.30 x 71		
Weight	lbs	2272.97		2837.35	2694.05	2976.24	3373

⁽²⁾ Considering a total system efficiency of 0.829 (Inverter 0.95 | Battery 0.9 | Wiring 0.97) DoD@90%

3.5 NAMEPLATE



A YANMAR COMPANY



QR CODE
Product
Documentation

EHR 45/60 V400/50HZ V4

Serial N°:		Weight:	1.350 Kg.
Manufacture Date:	31/01/2022	Dimensions:	1.62X1.15X1.92 m.
Rated Power:	31KW	Altitude Reference:	100 m.
Power Factor(cos PHI):	0,8	Temperature of ref.	25°C.
Storage Capacity:	56,8KWH.	LEP Battery Voltage:	48 VDC



221000659



DIRECTIVE 2014/35/EU
DIRECTIVE 2014/30/EU
MADE BY HIMOinsa

FACTORY: Ctra. Murcia - San Javier Km. 23.6 - 30730 SAN JAVIER (Murcia) Spain / Tel. +34 968 191 128

3.6 STANDARD ENVIRONMENTAL REFERENCE CONDITIONS

The environmental reference conditions for **EHR** are 25°C 100 msnm. **Reference conditions for all performance data included in this manual and data sheet**

Deterioration due to environmental operating conditions.

Include the reduction of the power of the inverter according to the delta calculated in the validation tests:

Inverters: continous output rating as function of temperature

In our datasheets inverters, and inverter function of Multis and Quattros, are rated at 25°C (75°F). On average, derating at higher temperatures is as shown below (see paragraph 4 for the theoretical background).

Temperature		Cont. output
°C	°F	%
Low temp.	100	
25	77	100
30	86	96
35	95	93
40	104	90
50	122	82
60	140	75
65	149	69
High temp.	0	

Operational limits.

When requesting an offer, Users/Clients must communicate any operating conditions that may affect the operation of the EHR. In addition to the environmental conditions indicated in the previous point, you should pay special attention to the characteristics of the loads you will connect, the power, the voltage and the power factor. You need to determine and indicate very precisely the load connection sequence.

- a. Power factor (fine body)
- b. Single-phase load.
- c. Starting of asynchronous engines.
- d. Low load profile.

Noise.

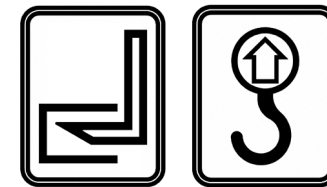
Information on the EHR sound protection level and the emission sound pressure level at the operator position is specific to each EHR model. This data can be found at:

- Weighted sound power level: consult this value in the CE declaration as well as in the marking of the equipment.
- Uncertainty: check with the manufacturer for each model.
- Sound pressure level: Check with the manufacturer for each model.

Sound power level measured in accordance with Directive 2000/14/EC, as amended by Directive 2005/88/EC.

4. UNLOADING, HANDLING AND TRANSPORT

- Once the EHR has been received, check that the material received corresponds to the delivery note, and that the merchandise is in perfect condition.
- To lift and transport the EHR, be sure to use lifting devices of sufficient capacity and follow the instructions set out in section 4.Unloading, handling and Transport. Any loose or pivoting parts must be securely fastened before lifting the equipment.
- When moving and especially when lifting the EHR, it is recommended to use the points specifically designed for these functions, previously verifying the optimal state of the lifting points.



- It is totally forbidden to use other retention points, located above the EHR.
- Any EHR that is damaged, for any reason, during transport, storage and / or assembly, must not be started-up before a check has been carried out by our specialised personnel.
- If you want to store the equipment until use, it is advisable to have a room properly protected from chemical agents that can deteriorate its components.
- Unpacking must be carried out carefully, avoiding causing damage to the material during such operation, in particular when levers or metal utensils are used.

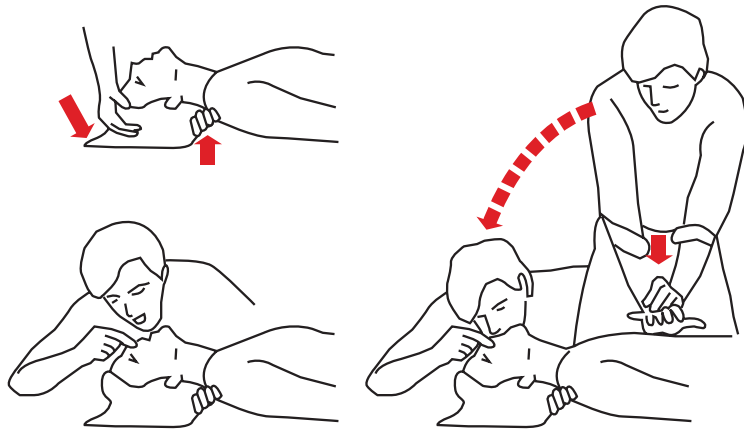
4.1 SAFETY DURING INSTALLATION AND INITIAL START-UP

- The installation of the EHR and its corresponding accessories must be carried out by specialised personnel. In case of any difficulty during the installation, please contact HIMOINSA's Technical Department.
- You must know the emergency procedures related to the installation to be performed and it is advisable to have a fire extinguisher in the vicinity of the EHR. You can check with the fire department for more information related to fire prevention.
- Always wear a protective helmet, footwear, safety gloves, goggles, and dry, close-fitting clothing.
- Do not modify the original protectors, located on all exposed rotating parts, hot surfaces, air intakes and live parts.
- Do not leave disassembled parts, tools or any accessories on the equipment, in its vicinity or inside the EHR.
- Never leave flammable liquids or flammable cloths soaked in liquid near the equipment (including lamps).
- Take all possible precautions to avoid risks of leakage. Ground the common point of the EHR designed to be grounded. Ensure that this grounding has been carried out in accordance with the relevant regulations. For more information, see Section 6.1.7 General Guidelines. Grounding.
- Place a "DO NOT MANOEUVRE" sign on all the circuit breakers that isolate the parts of the installation on which work is to be carried out.
- Install the necessary safety protections on the parts that complete the installation.
- Isolate all disconnected links and wires. Do not leave the EHR power terminals exposed.
- Check and ensure that electrical power links and auxiliary services are performed correctly.
- Make sure that the power cords you install meet the requirements of the relevant regulations, as the use of inappropriate cables can cause serious damage to both equipment and people (due to dangerous electrical conditions).
- Check that the equipment's shutdown devices are in perfect working order.
- Check that the ventilation is working properly - hot air from the inverters must be expelled into the atmosphere.
- Locate the position of all fire extinguishers and other protective and emergency devices and know how they work.

- Check that the equipment is clean, that the surrounding area and leakage routes are clean and unobstructed. Check for obstructions in openings or grooves.
- Check if personnel are working on other equipment in the area and if such work is hazardous and affects the operation of the facility.
- In the case of installations in environmental or operating conditions other than those designed by the EHR, consider the possible correction factors to be applied. This information can be found in the data sheet or on the identification plate of the equipment, section 5.3. Deterioration due to environmental operating conditions.

4.2 SAFETY DURING OPERATION

- Do not allow people who do not know the safety conditions, children or animals to access the operational area of the EHR.
- The person responsible for operating the EHR must remain alert and in the necessary conditions for an adequate response and interpretation. He or she must never operate it while psychically or mentally fatigued or under the influence of medication, drugs or alcohol.
- It is recommended that there be a minimum of two people during operations that may pose a health risk, especially electrical risks.
- Do not touch the EHR, especially the cables, copper terminals, and exposed connections, while the equipment is running, as they are live. In the case of electric shock, the first thing to do is to stop the equipment. If this is not possible, try to free the victim from the electrical power source using a non-conductive element. If the victim is partially or totally unconscious, perform cardiopulmonary resuscitation (CPR) and seek medical assistance immediately.



- Do not touch the equipment until the EHR is completely turned off.
- Never connect loads higher than the POWER RANGE of the EHR.
- The lines of the planned loads to which the generated power will be supplied must always be connected to the equipment before the start-up of the equipment.

- The power lines that power the equipment must be connected to the equipment before the equipment is put into operation.
- Stop the EHR immediately if you detect any type of abnormal operation, such as excessive vibrations, leaks, fumes, or loss of output power.
- Keep the body doors closed in case it is exposed to the outside. Otherwise, if it is located in a dry and clean place, you can leave the doors open.
- Maintain adequate ventilation to ensure that your EHR works properly. Without this ventilation, accidents or property damage could be caused.
- Safety labels should be kept clean and in places predetermined by the manufacturer.

4.3 SAFETY DURING MAINTENANCE

- Any checks and / or maintenance on the EHR must always be carried out by specialised personnel.
- Maintenance interventions should be performed with the EHR shut down. If the equipment is shut down after a period of operation, wait for it to cool down, be careful not to burn yourself, as some components may be extremely hot if the equipment has just been shut down.
- Before operating on any component of the electrical installation, turn off the equipment (See "6.3 Disconnection of storage equipment").
- Before opening the electrical panel, authorized personnel must disconnect the equipment (See "6.3 Disconnection of storage equipment").
- Periodically check both the tightening and insulation of the connections.
- Any operation and/or maintenance procedures not expressly indicated in the user manuals must be notified to the manufacturer for approval.
- Do not make modifications to the product without the knowledge and express authorisation of our Technical Department.
- Spare parts must correspond to the requirements defined by the manufacturer. Only use original spare parts. For spare parts, please only contact authorised spare parts distributors or workshops in the

- HIMOINSA support grid. For a correct determination of spare parts, always specify the data indicated on the equipment plate, and the type of EHR.
- Periodically monitor the status of the different components of the EHR.
- Periodically watch out for electrolyte leaks in the batteries.
- Do not regulate your equipment for a performance other than those provided by the manufacturer.
- Do not touch the power supply ducts when the equipment is in operation.
- When working on parts that may be live, always check that your hands and feet are dry. We recommend the use of insulating platforms when performing manoeuvres.
- Always keep the EHR clean, removing any stains.
- Do not perform tasks that require the presence of several people on your own, especially when operations must be performed on manoeuvring parts such as switches, disconnectors, fuses and / or live devices.

4.4 BATTERIES

- Periodically check the battery terminal connections to make sure they are clean, airtight, and weatherproof.
- Do not make any changes to the DC bus of the batteries. It is enough to check the DC bus to make sure the terminals are tight enough.
- Battery maintenance should be performed or supervised by personnel with knowledge of the batteries and the necessary precautions.
- When replacing batteries, make sure they are of the same type and number.
- Do not throw the batteries into a fire as they can explode. Do not open or mutilate the batteries as the electrolyte released is harmful to the skin and eyes.
- Batteries should only be replaced by the manufacturer or authorised personnel. Handling batteries or control systems can be dangerous and will invalidate the warranty.
- Batteries in some countries are considered hazardous waste. Use the appropriate containers or contact an institution responsible for the collection of this type of waste.

5. START-UP AND OPERATION OF THE EHR

The installation of the EHR must be carried out by qualified personnel, in accordance with the regulations in force in the country where the installation is to be carried out.

5.1 INSTALLATION: GENERAL INSTRUCTIONS

To carry out the installation, a series of general considerations must be taken into account, regardless of where the EHR is located. Complemented by the specific recommendations of each installation, shown in this section..

LOCATION OF THE EHR.

It should be verified that the EHR doors can be fully opened, that access for maintenance and overhauls is possible, and that the cooling system is working properly.

The location of the EHR is of utmost importance because the proximity of the power supply and its grounding point must be taken into account, proper ventilation (leaving at least one meter on all sides of the unit and making sure that the hot air flows that may be generated by the internal elements are directed away from the equipment). The unit should be placed upright on a flat, solid surface. Make sure the unit is not in danger of being submerged in water.

In general, the area where the EHR is installed must be properly fenced off to prevent access by unauthorised persons. Appropriate prohibition and danger signs should be placed in visible areas.

VENTILATION

The ventilation of the EHR plays a key role in its proper functioning and durability.

Improper ventilation can cause excessive temperatures around the EHR or inside the gensets, causing overheating and loss of operational efficiency.

Adequate ventilation should have the following characteristics:

The following premises must be taken into account:

- Heat from other sources should be considered in the design of the ventilation system.
- The design of the ventilation system of the equipment will be carried out with all doors closed.
- The EHR will be installed depending on the direction of the prevailing winds, both indoors and outdoors.

The altitude at which the genset is installed must be taken into account since the greater the altitude the lower the density of the air, requiring a greater flow of air than equipment located at sea level.

CONNECTING AND DISCONNECTING THE EQUIPMENT.

Explanation of the equipment connection:

1. Final protocol for the start-up of the equipment.
2. Physical connection steps.
3. Steps to follow to connect batteries, inverters.

The equipment is supplied ready to be connected to users' loads and power sources. When making connections, it is necessary to respect the conditions indicated in the diagrams that are supplied with the equipment.

The choice and sizing of the cables is the responsibility of the installer who performs the installation, depending on the type of cable and the current regulations that are applicable in the country in which the installation is carried out.

Power cables must be connected to the line terminals located at the bottom of the electrical panel, and must be located in suitable conduits, tunnels or cubicles carrying protective conductors. It is recommended that cables of different voltages be separated by a minimum distance between layers of 25 cm, always placing those with the highest voltage in the deepest area, thus avoiding possible magnetic interference.

GROUNDING (TNS, IT).

Differentiate the TNS, IT for the connection of the equipment and the behaviour of the equipment with regard to the different grounding systems.

Metal parts of installations that are exposed to contact with people, due to an insulation defect or accidental causes, could be live. To ensure the protection of

people, the electrical installation and the equipment, the customer must ground the EHR.

To carry out grounding, the EHR contains a main grounding terminal. The customer shall connect his grounding rod to the grounding installation of the equipment by means of an insulated copper conductor with a minimum section of 16 mm² or by means of a bare copper conductor with a minimum section of 25 mm².

The materials, dimensions and depth of the grounding rods must be chosen in such a way that they resist corrosion and present adequate mechanical resistance. They should be installed vertically in the ground. The resistance of a grounding rod depends on its dimensions, its shape and the resistivity of the soil in which it is embedded. This resistivity is usually variable from place to place and varies with depth.

The choice and sizing of the grounding conductors and rods will be the responsibility of the installer who performs the installation, taking into account the local and national regulations applicable in their jurisdiction.

OUTDOOR INSTALLATIONS.

After consulting the general installation instructions, shown in the previous section, and the corresponding regulations, if the EHR is being installed outside, special attention should be paid to the noise generated, the environmental conditions and the influence of seasonal and meteorological changes on the characteristics of the soil and the environment.

5.2 PRE-START-UP CHECKS

These operations should be performed in the following situations:

- Before start-up.
- After the installation of the equipment.
- After a general overhaul.
- After performing maintenance operations.
- After a long period of inactivity.

5.3 START-UP

Perform all the checks described in the previous sections. Verify that the magnetothermal and differential switches are in the OFF position.

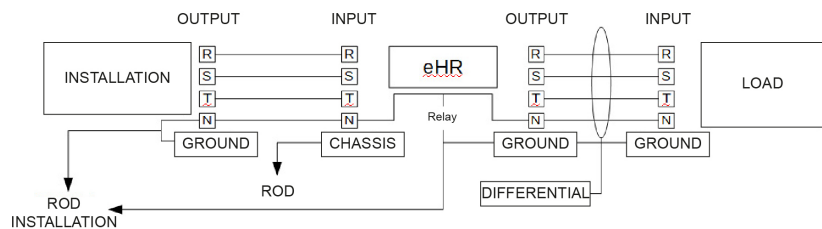
NOTE:

It is not recommended that batteries be discharged below 17%, as the unit will enter standby mode and the procedure described in section 6.1.1 will have to be applied.

Check the grounding connections of the system. Below we show the most common grounding system with the connections that should be taken into account:

TT GROUNDING SYSTEM.

Electrical distribution grid as primary power supply

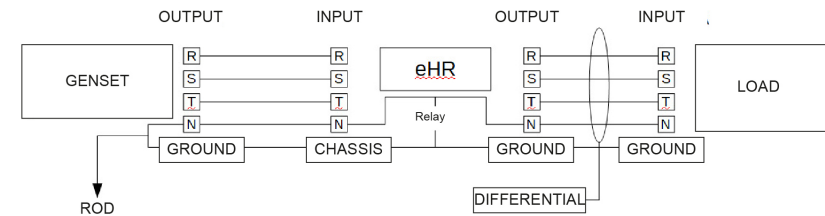


As per the above diagram, the steps to follow to perform a TT grounding are as follows:

1. The grounding of the electrical distribution grid must be bridged with the neutral.
2. The grounding of the electrical distribution grid and the grounding of the EHR must be attached and wired to a grounding rod (Rod # 1)
3. The EHR chassis must be connected to an insulated grounding rod (Rod #2)
4. The plate that attaches the chassis to the busbar must be completely clean.

TNS SYSTEM

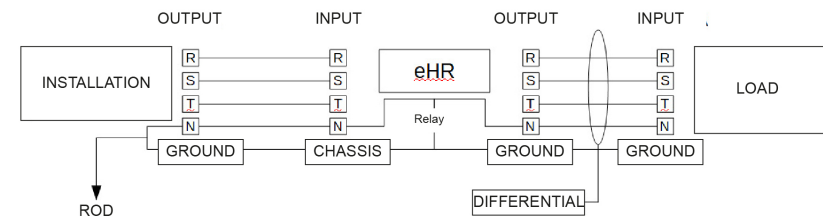
Generator set as primary power supply



As per the above diagram, the steps to follow to perform a TNS grounding are as follows:

1. The ground of the generator set (neutral) and the chassis must be bridged.
2. The ground of the generator set (neutral) and the ground of the EHR must be connected and wired to a grounding rod.
3. The EHR chassis and the EHR ground must be bridged.
4. There must be a plate connecting the EHR chassis and the EHR ground node.

Electrical distribution grid as primary power supply



As per the above diagram, the steps to follow to perform a TNS grounding are as follows:

1. The grounding of the electrical distribution grid must be bridged with the neutral.
2. The grounding of the electrical distribution grid and the grounding of the EHR must be attached and wired to a grounding rod (Rod # 1)

1. The EHR chassis and the EHR ground must be bridged. There must be a plate connecting the EHR chassis and the grounding busbars.

The EHR equipment incorporates two input channels, which are described below:

Primary power supply input

This input is intended for the connection of a generator set or an electrical distribution grid (three-phase), feeding up to a maximum of 100A per phase. This input can be made in two ways, by busbar or by “SOCKET” connection. To indicate the chosen selection, the selector included in the front panel must be changed to the correct position. In the following image we can see the selector and the position corresponds to each of the connection modes.



In the panel the selector is identified by Sockets or Busbar

Maintenance input

This input is intended to charge the batteries externally, using a conventional single-phase power supply from a house. This input powers the batteries up to a maximum of 16A.

To make a connection from a primary power supply you must abide by the following connection guide, keeping in mind that the phase sequence must be respected.

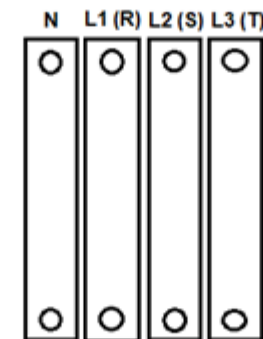
TERMINAL CONNECTION

Phase arrangement

When distributing the phases in the busbar, the arrangement of each of them must be taken into account, since an incorrect phase sequence can set off an alarm and cause the system to shut down.

When wiring a generator set, the user will not have any problem with the connection of the phases, since in the output of the generator sets the phase sequence is indicated, so each phase must be connected respectively. In the opposite case, when you want to wire an electrical distribution grid, it must be taken into account that the distribution of phases is not known and that for this reason, the phases must be changed until the EHR does not show the “*Incorrect phase rotation*” alarm.

An image is provided below that indicates the distribution of phases that must be wired in the input busbar (busbar on the left)



Once the busbar is wired, you must proceed as indicated in the next section to perform the input connection routine.

Steps to follow

- Make sure that the AC source to be connected to the unit is turned off.
- On the HICORE's built-in display, verify that no operating mode is

- being performed and that the inverters are turned off. Otherwise, shut down the operating mode being performed and verify that the inverters are turned off.
- Place the selector in the “ON -” position, see 3.2 Identification of the main components.
- Open the lower distribution door, see 3.2 Identification of the main components.
- The busbar input connection is on the left. Make sure the supply terminals are dry, clean any excess moisture with an absorbent cloth.
- Connect the AC power supply connectors to the input busbar and make the connection as explained in the “Phase Arrangement” section above
- Close the lower distribution door.
- If the AC power supply is a generator set with automatic start-up, connect the generator set start-up control input to the start-up relay connector that starts the EHR, see 3.2 Identification of the main components.
- Select an operating mode in the HICORE to start the inverters.
- If the AC power supply is a generator set without automatic start-up or an electrical distribution grid, start or connect the AC source.

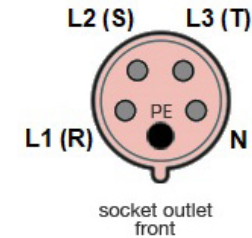
If the AC power supply is an electrical distribution grid, you should keep in mind that as the phase distribution is not known, the phases should therefore be interchanged until the EHR does not show the “Incorrect phase rotation” error.

SOCKET CONNECTION

Phase arrangement

As mentioned above, since the EHR equipment incorporates a power input through the “” connector, this input allows the connection of the primary power supply quickly. Just as the connection by busbar must take into account the distribution of the phases, when connecting to an elective generator set, the user will not have any problems as the phases are marked and the sequence is known. If you want to wire the electrical distribution grid, it should be borne in mind that as the phase distribution is not known, the phases must therefore be interchanged until the EHR does not show the “Incorrect phase rotation” error.

An image is provided below that indicates the distribution of the phases to be wired in the input “SOCKET”.



Once the “SOCKET” is wired, the input must be connected to the EHR, and the steps described below must be followed.

Steps to follow

- Make sure that the AC source to be connected to the unit is turned off.
- The HICORE’s built-in display should check that no operating mode is being performed and that the inverters are turned off. Otherwise, stop the operating mode being performed and verify that the inverters are turned off.
- Place the selector in the “ON =” position.
- Connect the “SOCKET” connector of the AC power supply to the three-phase “SOCKET” input and make the connection as explained in the “Phase Arrangement” section above.
- If the AC power supply is a generator set with automatic start-up, connect the generator set start-up control input to the starter relay connector incorporated in the EHR, see 5.6. *EHR control panel*.
- Select an operating mode in the HICORE to start the inverters.
- If the AC power supply is a generator set without automatic start-up or an electrical distribution grid, start or connect the AC source.
- If the AC power supply is the electrical distribution grid, it should be noted that as the phase distribution is not known, the phases should be interchanged until the EHR does not show the “Incorrect phase rotation” error.
- Connect the maintenance AC power supply connector to the single-phase “SOCKET” input and make the connection.
- As soon as you make the connection of the EHR equipment, it automatically detects the operating mode (if it is operational) and only performs the maintenance charge.

CONNECTION OF DC INPUTS (SOLAR)

It is advisable to make the connection of the solar connectors when the equipment is turned off or when it is not performing any operating mode. Remember that the “array” of solar panels for each of the connectors that you integrate into the EHR must have the following limiting characteristics

Open Circuit Voltage (Voc) [V]	240
Short circuit current (Isc) [A]	23

EHR models include a maximum of 6 solar inputs (depending on the model). Solar panels connected to each of these inputs must meet the above specifications. If these limits are not respected and solar panels with much higher specifications are installed, the EHR equipment may suffer the risk of fire.

5.3.1. CONNECTING TO A GENERATOR SET.

A remote generator start-up function is provided to automatically control a generator set. The start and stop conditions are automatically performed in each of the operating modes, depending on the source intended to power the load.

***NOTE:**

The minimum capacity accepted by the system will be 10kw per genset

Connecting the automatic start-up of a generator set.

The generator set's remote start-up terminals are located on the lower door. This is the connection that is used to send a start or stop signal to a generator set using its automatic start-up cable. EHRs include these connectors so that a proper connection is made with whatever genset the user owns.

When setting up a hybrid system, the overall performance of combining an EHR with a generator set can be improved by making some simple adjustments to the type of generator set.

In auto-start mode, some types of generator sets start running as soon as a start signal is received. Others, however, have a number of delays that can prevent a diesel generator set from starting up and generating power as quickly as possible. These delays should be minimised whenever possible. Otherwise, the operation of the operating modes will not be 100% effective.

COMMON CAUSES OF GENERATOR START-UP DELAYS.

Delay in start-up.

This delay allows for brief “false start” signals and can last up to five seconds. When used with a hybrid unit, it is important that the diesel engine starts immediately. Whenever possible, this delay should be eliminated.

Preheating delay, safety ignition timer and heating timer.

Always try to reduce delays to the minimum accepted level.

AUTOMATIC MODE

Make sure the generator is in automatic mode.

5.3.2. CONNECTING THE AC OUTPUT

Connecting the AC output to the terminal

- Make sure that the AC source that is connected to the unit is turned off.
- On the integrated HICORE display, verify that no operating mode is being performed and that the inverters are turned off. Otherwise, stop the operating mode being performed and verify that the inverters are turned off.
- Open the lower distribution door see 3.2 Identification of the main components.
- If the AC power supply is a generator set with automatic start-up, disconnect the generator set start-up control input from the starter relay connector incorporated in the EHR, see 3.2 Identification of the main components.
- The bus output bar is on the right. Make sure the supply terminals are dry, clean any excess moisture with an absorbent cloth.
- Connect the AC output charging connectors to the output terminals and make the connection
- If the AC power supply is a generator set with automatic start-up, connect the generator set start-up control input to the start-up relay connector that incorporates the EHR, see 3.2 Identification of the main components.

- Close the lower distribution door.
- Select an operating mode in the HICORE to start the inverters.
- If the AC power supply is a generator set without automatic start-up or an electrical distribution grid, start or connect the AC source.

Connecting the AC output using bases.

- Make sure that the AC source that is connected to the unit is turned off.
- On the HICORE's built-in display, verify that no operating mode is being performed and that the inverters are turned off. Otherwise, stop the operating mode being performed and verify that the inverters are turned off.
- If the AC power supply is a generator set with automatic start-up, disconnect the generator set start-up control input from the starter relay connector incorporated in the EHR, see 3.2 Identification of the main components.
- Connect the AC output load connector to the desired output of the EHR baseboard.
- If the AC power supply is a generator set with automatic start-up, connect the generator set start-up control input to the starter relay connector incorporated in the EHR, 3.2 Identification of the main components.
- Select an operating mode on the HICORE for the inverters to start.
- If the AC power supply is a generator set without automatic start-up or an electrical distribution grid, start or connect the AC source.

The routine of starting and shutting down the equipment should be carried out at specific times when the user deems it convenient. The current arrangement of EHR equipment is made so as not to have to perform this routine, since the battery panel is difficult to access, which means that this routine must be performed by the technicians who operate the machine or the owners of the EHR equipment themselves.

Once all connections are completed, the unit is ready to perform a full start-up. Proceed as follows.

Before you start

It must be taken into account that before carrying out a start-up you must make sure that the machine is in a state of shutdown, as if this is not the case, the start-up routine will not proceed properly and may result in a possible failure during start-up. Attached below is the pre-arrangement of the EHR equipment to perform a proper start-up.

- All inverter switches should be set to "OFF".
- All battery switches should be set to "OFF".
- All inverter battery switched should be set to "ON"
- The DC circuit breaker should be set to "OFF"

During start-up

Once you have checked that the status of the machine is correct, you can proceed to the complete start-up of the EHR equipment. To do so, you must follow the following steps carefully, respecting the indicated order:

1. Set the switches of all batteries to "ON".
2. Press battery button number 1 (red button)
3. Wait 2 minutes.
4. Set the DC circuit breaker to "ON".
5. Set the inverter battery switches to "ON"

For more detailed information, see *Start and stop procedure for the installer*.

Whenever you want to store the equipment for a long time or want to perform a complete stop to restart the equipment or for reasons, you should follow these steps carefully and following the order indicated:

1. Set the DC circuit breaker to "OFF"
2. Set the switches of all batteries to "OFF"
3. Set the switches of all inverters to "OFF"
4. If maintenance is to be performed, set the inverter battery switches to "OFF"

For more detailed information, see *Start and stop procedure for the installer*.

To restart the EHR plant, you need to proceed as follows:

1. Go to the settings screen and tap “VE. Reset”
2. Then press the Start button for about 10 seconds
3. When it is observed that the equipment has been completely turned off, press the start button again for about 10 seconds.
4. When you see that the control panel has started up, the equipment will restart completely.

If an emergency occurs and the unit needs to be turned off, press the emergency stop button located on the top of the EHR equipment, see 5.6. *EHR control panel*. The unit shuts down immediately. To reset the EHR equipment, the emergency stop button must be disengaged by turning it clockwise.

WARNING:

As the activation of the emergency button causes the equipment to shut down, when an emergency stop is real, the “Emergency shutdown” alarm and “Devices off” will appear at the same time.

Once the issue has been fixed, follow the instructions in *section 8. Start-up and operation of the EHR..*

5.4 CHECKS AFTER THE START-UP OF THE EHR

The following checks must be carried out:

- Electrical checks (voltage, intensity, frequency, etc.).
- Mechanical checks (tightening screws, sockets, etc.).
- Safety checks (emergency stop, magnetothermal switch, differential switch, etc.).

5.5 START-UP SEQUENCE FOR EHR

EHRs are equipped with an electronic protection and control centre, the characteristics of which are defined in the appropriate sections (See “7.5. Quick Guide to Using the EHR Plant”). Also included is a protective dashboard (fuses, magnetothermal switches, differentials...), voltage switch elements and input/output connection terminals

In an EHR plant, operations and equipment operating statuses are displayed with a set of light, text, and graphic indicators. For information on these and other instructions and operations, please refer to the relevant documentation at EHR headquarters.

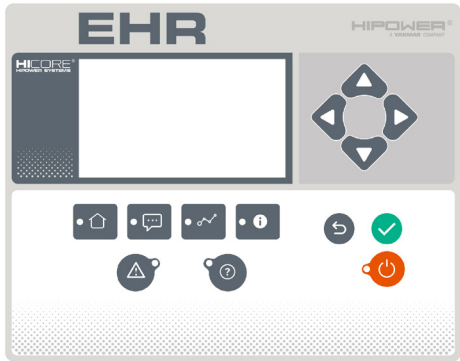
When a new unit is shipped, it is completely disconnected for safety reasons. When starting the unit up for the first time, **the following steps must be followed in the correct order.**

1. Check that all battery-inverter insulators (rotary switches located under the inverters) are in the “OFF” position (note that, depending on the configuration of the unit, the insulators can be accessed from either of the side doors).
2. Check that the switches on the front of each inverter are in the “OFF” position.
3. The control panel switch, located on the control panel must be in the “OFF” position.
4. Open the two lower side doors covering the battery section.
5. Set the battery-inverter insulators (rotary switches located under the inverters) to “ON” and wait 20 seconds. You will need the key for this.
6. Press the AC on/off switch that turns one of the batteries on. This switch is located on each of the battery modules used to turn on all the batteries in the unit.
7. Press the red button on battery number one for several seconds (this will be indicated).
8. Wait approximately 2 minutes until all battery status LEDs are green. (They don’t have to be steady; they can flash at random to signal communication between batteries.)
9. Set the switches on the front panel of each inverter to “ON”
10. On the control panel, turn the switch to “ON” to initialise the control panel.
11. Close the battery compartment and inverter access doors.
12. If the SoC of the unit is less than 90%, the batteries will be in standby mode and therefore the control panel will not start. Refer to the manual to perform a maintenance charge to stabilise the batteries at 100% of the SoC. See section 5.9. Emergency battery charging (SoC <10%).
13. The unit is now initialised. Refer to the user manual for instructions on how to configure and operate it instructions.

5.6 HICORE CONTROL UNIT

FRONT OF THE DISPLAY MODULE

The *display module* has a backlit screen and different LEDs to monitor the status of the plant. It also has push buttons that allow the user to command and programme the control panel.



EHR Display Module

1. Backlit screen with 4 lines by 20 digits.

NOTE:

The screen goes into low power mode (no backlight) after 2 minutes after no keystrokes are detected.

2. Central buttons

- Navigation buttons
- Help and alarm buttons
- Power button
- Main screen button
- Application button
- Log button
- Information and settings button

3. Status LED.

- SUPPORT INDICATION LED
- ALARM AND ADVERTISING INDICATION LED
- POWER INDICATION LED
- CURRENT DISPLAY INDICATION LED

Buttons on the control unit.

1. Central command buttons

	Push button to turn on the screen.
	Menu. Display of the equipment menu on the screen.
	Applications. On-screen display of the operating modes of the equipment.
	Graph. Generates graphs of the equipment log for the last 8, 12 and 24 hours.

	System status. Displays the status and settings of the equipment on the screen.
	Alarms.
	Access help. During navigation, when the LED is illuminated, you can access the help menu.

2. Navigation buttons

	Validate. Enter the menus and validate the data entered.
	Right (►). Move the selection to the right of the screen.
	Left (◄). Move the selection to the left of the screen.
	Go up (▲). Move the selection up the screen and increase the programming values.
	Go down (▼). Move the selection down the screen and decrease the programming values.
	Return to the previous screen.

5.7 ACCESS PASSWORD

SECURITY LEVELS

HICORE has different access levels for different users, as shown below:

Level 1	Level 2
User	Rental
1111	1285
- Navigation - Start and stop	- Calendar - Date/Time - Language - IP/CCLAN - Delete partial history

***IMPORTANT:**

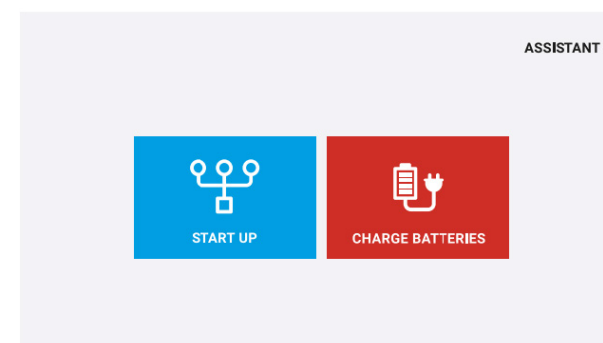
once accessed with a user level, this level will remain active until the work mode is restarted with the accesses available to all users.

5.8 HICORE NAVIGATION

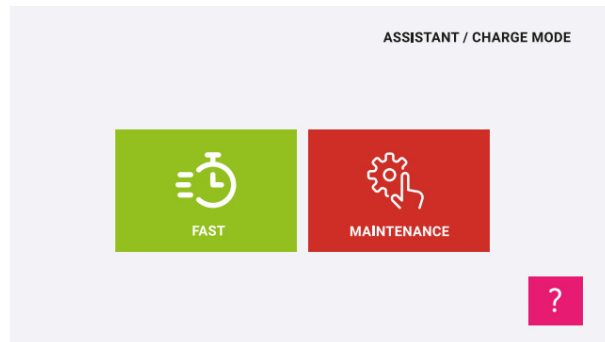
As a first step, once the EHR has been started, the control panel will suggest that you access the configurator.

Using the “wizard”, you can configure the unit for the desired working mode.

First of all, you need to indicate what action you want to perform on the system, either to perform a load, or run an operating mode.

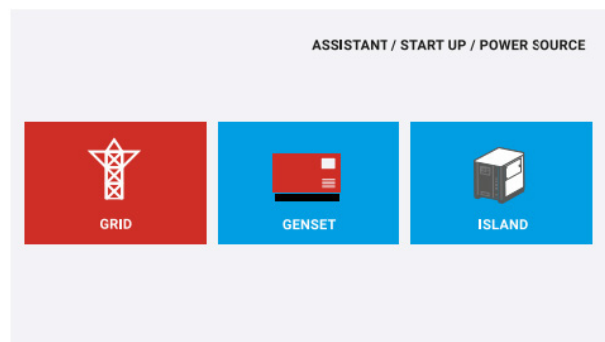


If you want to “CHARGE THE BATTERIES” you must indicate the type of charge to be used, whether a fast charge or a maintenance charge.

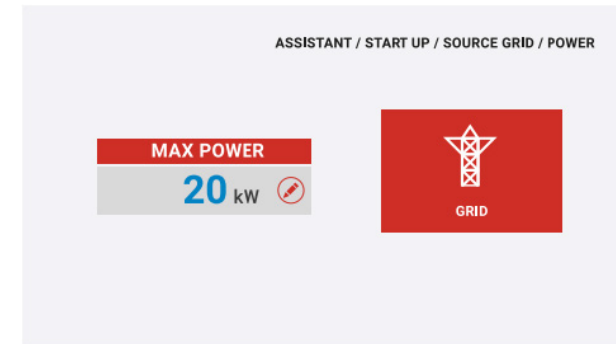


5.9 CHARGE BATTERIES

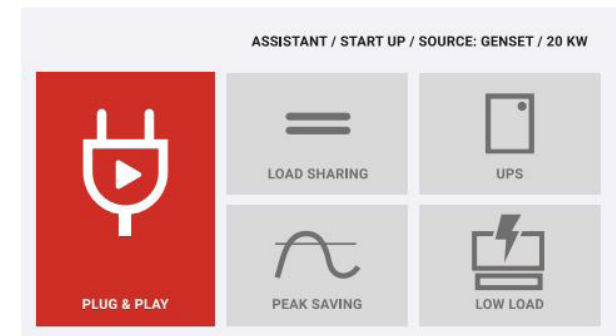
If you have selected “START UP” you have to indicate what the system’s primary power source is, whether a generator set or an electrical distribution grid or without an energy source (ISLAND).



Once you have indicated the primary power source, the maximum power of that source will be configured. This data will allow the EHR to control the system to improve the efficiency of the primary energy source and increase its useful life as in the case of generator sets. If it has been previously selected that the EHR is not a power supply (ISLAND) it will not be necessary to configure the maximum power of the primary power source, as it does not exist.



Next, we will select the desired mode of operation: the description of the modes of operation are indicated in 3.2. *Operating modes*.



If no primary power supply (ISLA) has been chosen beforehand, the only available operating mode will be “PLUG & PLAY” because it is not possible to intelligently combine battery charging and charging power using a primary power supply.

5.10 WORKING MODES

The EHR integrates several operating modes aimed at adapting to all the work scenarios that may occur in the electrical system, allowing greater performance and greater adaptability of the equipment currently available in the installations.

	PLUG & PLAY mode. The use of the battery and the generator set or grid is enabled dynamically.
	UPLOAD SHARE mode. Grid usage is enabled at the same time as battery usage.
	Power Booster mode. Only available for genset connection. Working mode designed for starting electric engines.
	UPS mode. Only available for grid connection. When the primary power supply is disconnected, the battery will be connected.
	PEAK SAVING mode. The generator set or the grid is enabled up to a single power. Beyond this power, any additional energy needed will be delivered by the battery.

	LOW LOADS mode. The battery is enabled and beyond a certain power, the battery is disconnected and the generator or the grid is connected.
---	--

To select each of these modes, you must select the one you want to use. Next, you have to indicate the corresponding configuration data and finally press “START”. If we want to finish this operation we must click on the “ASSISTANT “ button that will take us to the tab of the operating mode that is currently running and all we will have to do is press “STOP”.

All operating modes include a help function to advise the user at all times. To access this content, you just have to click on the “HELP” button, whose status LED will have started flashing beforehand to indicate that help is available.



WARNING:
If you are charging a battery, it will not be possible to activate the operating modes, the charging must be disabled and then you can activate the desired operating mode.

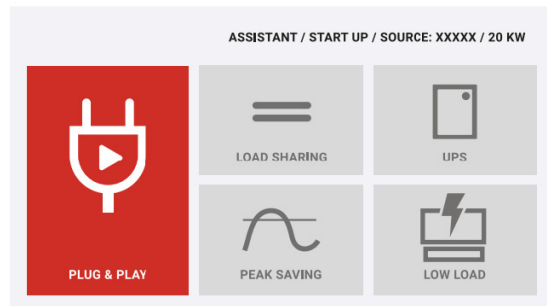
WARNING:
Remember that to change the operating mode it will be necessary to press “STOP” to halt the operating mode that is currently being used.

WARNING:
In all hybrid working modes that combine the grid / generator set with the batteries, the minimum power delivered by the batteries in any case will be 5Kw.

WARNING:
During the charging process in each working mode, the charge will be limited to the absorption phase of the battery; this means that the battery will be considered fully charged around 97%. To make a 100% charge you must access the battery charging menu.

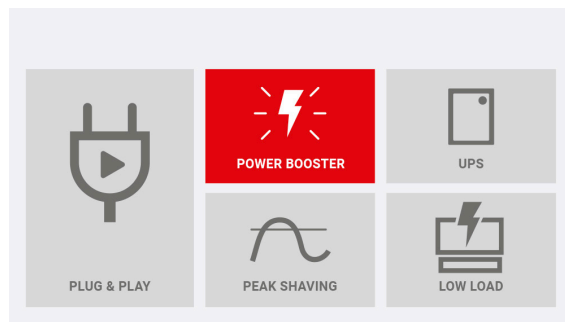
Depending on which primary input source you select, the available applications will change:

5.10.1. ELECTRICITY GRID AS AN EXTERNAL SOURCE:



5.10.2. GENERATOR SETS AS AN EXTERNAL SOURCE:

When working with a generator set, a specific function appears to start the electric engine.



5.10.3. PLUG AND PLAY

“PLUG & PLAY” is the default mode for the EHR equipment. The intention is to offer the user a way that avoids having to insert configuration parameters and it incorporates a wide functionality to adapt to most applications. This mode depends on the previous configuration that has been made on the equipment, distinguishing between two functionalities:

When the electricity distribution grid or a generator set has been selected as the primary energy source.

In this case the “PLUG & PLAY” mode will act intelligently by combining three operating modes:

- LOW LOADS
- UPS
- PEAK SAVING

Thus, for an initial moment in which the loads present at the output are low, the equipment will operate in “LOW LOADS” mode, until the threshold defined by default is exceeded. This threshold is defined as 30% of the maximum power that the generator set or the electricity supply grid can supply. If it is working without any load at all, the EHR will take advantage of this moment to charge the batteries. Once this threshold has been crossed, the generator set will start up or the grid input will be enabled to power the output and charge the batteries if necessary. It should be noted that the genset will continue to work only with the grid or generator set until the maximum performance power of the generator set is reached, which will be given as 90% of the maximum power that the generator set can supply. If the grid is chosen, this threshold will be given by the maximum power selected by the user. Once this power is reached, the “PEAK SAVING” function will be performed, which means that the power of the generator set or grid will be limited to that value while the peaks present at the output will be powered by batteries or by work batteries and the generator set or the grid together.

When no primary power supply has been selected, the system will operate in island mode

In this case, the “PLUG & PLAY” mode will only work with batteries, supplying the load by batteries and if using renewable energies, it will charge the batteries and at the same time discharge them to power the load. This mode is recommended for specific uses with very low consumption as this will be when the system

stops due to a low state of charge.

In any case, to run this mode of operation you just have to select it and then press “START”.



WARNING:

When the battery SoC falls below 20%, the EHR will operate at 100% with the generator set or grid until the charging stage is completed. Once the batteries are fully charged, the equipment returns to normal operation. This is the case if we are working in “LOW LOADS” mode. When this condition occurs during “PEAK SAVING” or “UPS” mode, the equipment will shut down in the “PLUG & PLAY” mode.

WARNING:

Whenever the generator set is turned on or the power grid is enabled, there will be a minimum battery-only operating time until the inverters synchronise with the wave from the connected source.

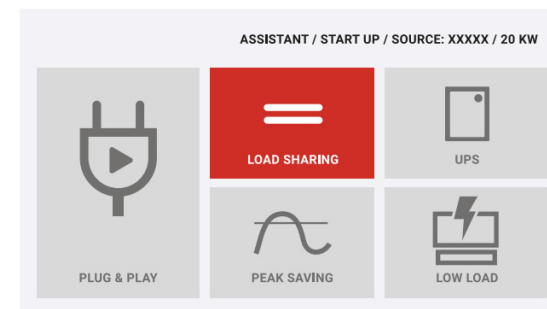
WARNING:

When starting this operating mode, there will be a delay of [5 – 30] s until the EHR output is enabled.

RECOMMENDATION:

When a mode shutdown occurs due to an unwanted SoC (<20%), it is necessary to fully charge the batteries externally using maintenance or fast charging. This value may vary slightly by up to 17% by the hysteresis programmed into the system.

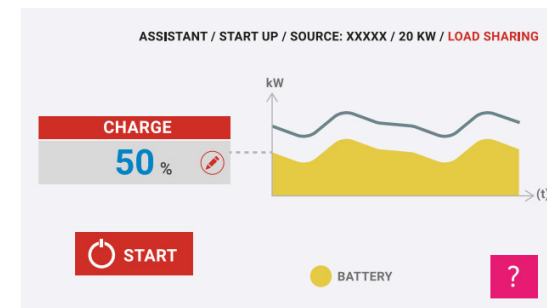
5.10.4. LOAD SHARING.



This mode is intended to allow energy sources to work in parallel exclusively with the power grid, as well as the battery. To achieve this function, the user must indicate the percentage (%) of energy that he or she wants to feed through the grid, since the rest of the energy will be provided by the battery.

Once this configuration is made, the system will adapt by feeding the load present at the output, with the proportion chosen by the user. This mode of operation is useful to take advantage of solar energy, since on sunny days with the usual demand for energy, the load present at the output can be fed with a proportion of generator set or grid and the rest will be provided by solar energy (if it is greater than the demand, it will charge the batteries and if it is lower, the battery will provide the missing power). When the maximum threshold of supply power, configured by the user, is exceeded, the input will be limited to that value and the rest of the energy will be provided by the battery.

To execute this mode of operation it is necessary to insert the percentage (%) of energy contributed by the grid and then press “START”. Below we can see some images that show us what this screen looks like.



WARNING:

In all EHR models, this is a minimum grid supply value. This minimum value depends on the chosen EHR model, see 4.6. *Electrical assistance*. The EHR will ignore the percentage selected by the user until that threshold is exceeded.

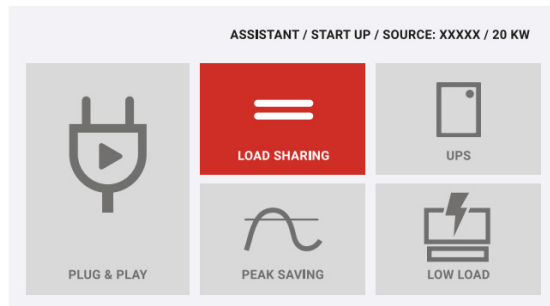
WARNING:

When the battery SoC drops below 20%, the equipment operates with a 100% percentage of the grid until the charging stage is complete. Once the charging is complete, the equipment returns to normal operation.

WARNING:

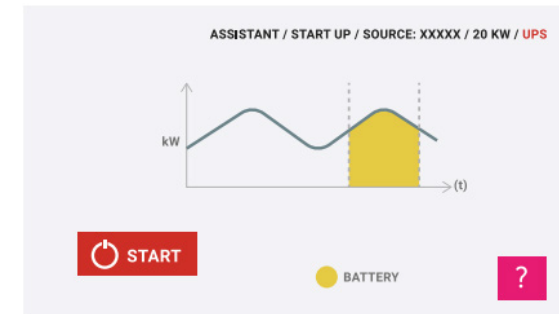
When starting this operating mode, there will be a delay of [5 – 30] s until the EHR output is enabled.

5.10.5. UPS



This mode is intended to achieve uninterrupted power at the output. To do this, the EHR equipment supplies the loads through a generator set or grid and if the primary power supply disappears, the charge is instantly supplied by the batteries. When the maximum supply power threshold set by the user is exceeded, the input will be limited to that power value and the batteries will provide the missing power.

To run this mode of operation you just have to select it and then press “START”.



WARNING:

When the battery SoC decreases by 20%, the equipment will perform a “UPS” mode shutdown.

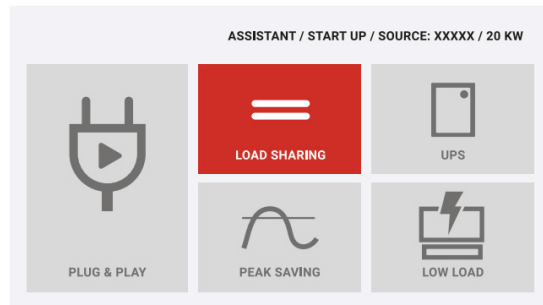
WARNING:

When starting this operating mode, there will be a delay of [5 – 30] s until the EHR output is enabled.

RECOMMENDATION:

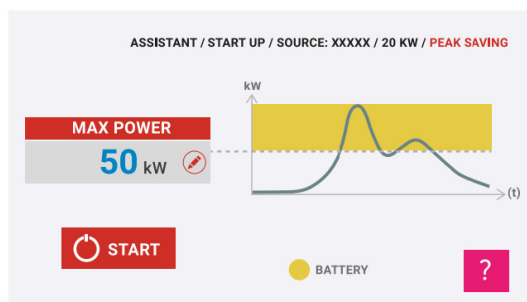
When a mode shutdown occurs due to an unwanted SoC (<20%), it is necessary to fully charge the batteries externally using maintenance or power input.

5.10.6. PEAK SAVING



This mode is intended to feed the loads present at the output only by a generator set or grid, except when there are large energy demands, higher than those that can be provided by the generator set or grid. In this case, a limitation on the input is established and the demand for extra energy is delivered by the batteries. You can set the maximum desired power to be provided by the generator set or the grid (this value will be limited to that indicated as the maximum power of the generator set or electrical grid, since this power must never be exceeded). It is important to size the generator set to avoid oscillations, being the minimum group power of the same capacity of the EHR to be used (recommended 30% higher) and perform the PEAK SAVING from 90% of its nominal power. If the value is lower, the system will indicate that the connected group is not powerful enough for this application.

To run this mode of operation it is necessary to insert the maximum power that will be provided by the generator set or the grid and then press “START”. Below we can see some images that show us what this screen looks like.



WARNING:

When the battery SoC decreases by 20%, the equipment exits “PEAK SAVING” mode.

WARNING:

When starting this operating mode there will be a delay of [5 – 30]s until the EHR output is enabled.

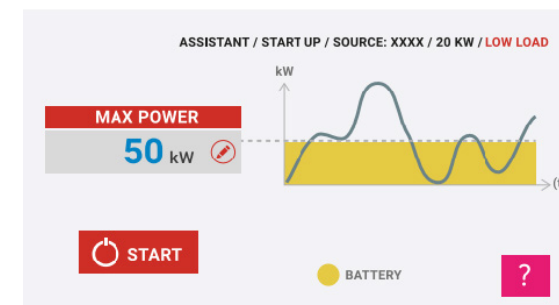
RECOMMENDATION:

When performing a mode shutdown due to an unwanted SoC (<20%), it is necessary to fully charge the batteries externally using maintenance or power input.

5.10.7. LOW LOADS

This mode is intended to power low loads only by battery, except when the energy demand exceeds a power threshold determined by the user, in which case the generator set is started up or the grid is enabled to work fully with this power source. When the maximum power threshold, set by the user, is exceeded, the input will be limited to this value and additional power will be provided by the batteries.

To run this mode of operation it is necessary to insert the maximum power at which it is powered by batteries and then press “START”. Below we can see some images that show us what this screen looks like.



WARNING:

When the battery SoC is 20%, the equipment will operate at 100% with the generator set or grid until the charging stage is completed. Once the charging is complete, the equipment returns to normal operation.

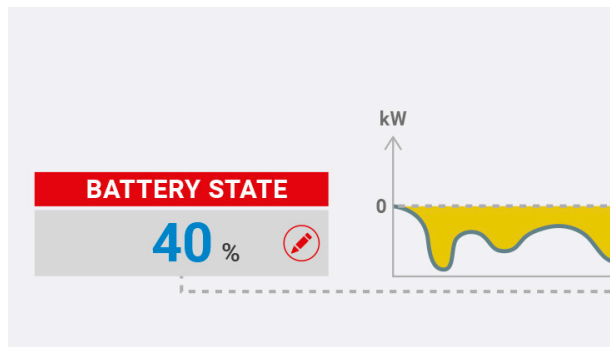
WARNING:

Whenever the generator set is turned on or the power grid is enabled, there will be a minimum operating time.

5.10.8. POWER BOOSTER

In this working mode, we take advantage of the overload capacity of the batteries to absorb the power peaks in the load. We configure the minimum state of charge level so that, beyond this level, the generator set starts up.

Once you have selected the level of charge status of the **SoC** batteries with the help of the cursors () and once you have made the connections, don't forget to press "START".



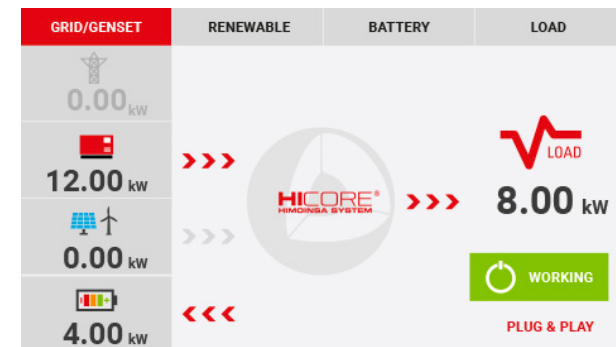
NAVIGATION

The initial tab allows users to visualise the most important information and thus be able to observe the power flows in a clear and graphic way.

This screen is the default for the EHR equipment and whenever possible the equipment will display it on the screen. If you want to access it quickly, you just need to press the "main menu" button.



In the main menu you can see what energy sources are available, how they are distributed and the amount of energy that is being produced and consumed. As you can see, the central graphic, represented as "HICORE", refers to the control unit of the EHR which is responsible for managing all the operations and scenarios that the EHR will encounter.



As you can see in the picture, we have three power supplies here, namely:

- Primary power supply, either the electrical distribution grid or a generator set.
- Renewable energies, currently only input from solar panels is available.
- This power supply will therefore always be available until it runs out, as it is integrated into the EHR equipment.

	Enabling AC input as an electrical distribution grid.
	Enabling AC input as a generator set.
	INPUT of DC as solar panels (PV).

To the right of the image, the installation to be supplied or “loaded” appears and at the bottom we can see the mode of operation that is running at that time with the indication of “WORKING”. If the system is idle, the “STAND BY” indicator will appear.

At the top of this screen, you can see selectable blocks indicated as:

- GRID/GENERATOR SET
- RENEWABLE
- BATTERY
- LOAD

GRID/GENSET	RENEWABLE	BATTERY	LOAD
-------------	-----------	---------	------

By selecting these boxes, it will be possible to see in detail the parameters of each of the available energy sources or the data of the installation to be supplied. Each of these tabs is explained in more detail below.

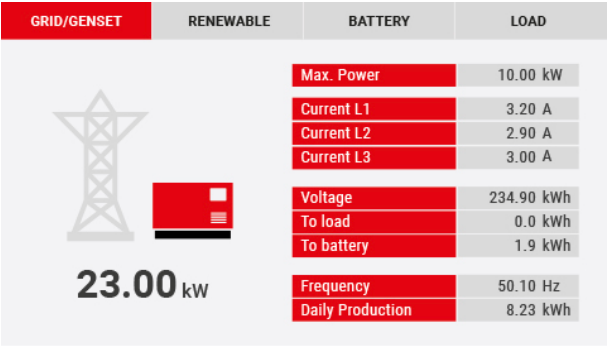
This screen shows the information generated by the electrical distribution grid or the generator set, depending on the primary energy source that has been selected beforehand. In this tab you can see the following data:

- Instantaneous power (kW)

- Maximum input power configured (kW)
- L1 phase current (A)
- L2 phase current (A)
- L3 phase current (A)
- Voltage (V)
- Power directed towards the “load” (kW)
- Power directed towards the battery (kW)
- Frequency (Hz)
- Energy produced daily (kWh)

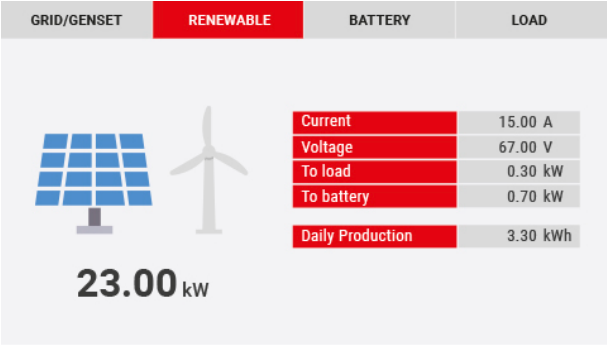
GRID/GENSET	RENEWABLE	BATTERY	LOAD
			
23.00 kW			
Max. Power			10.00 kW
Current L1			3.20 A
Current L2			2.90 A
Current L3			3.00 A
Voltage			234.90 V
To load			0.0 kW
To battery			1.9 kW
Frequency			50.10 Hz
Daily Production			8.23 kWh

If you have selected the generator set option, the data to be displayed will be the same, although HICORE will make the appropriate adjustments to allow for a power supply such as a generator set.



In this screen you can see the information generated by the installed solar panels. We can see the following data:

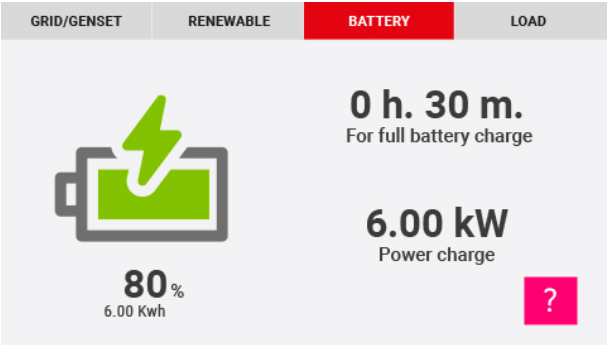
- Instantaneous power (kW)
- Solar panel current (A)
- -Solar panel voltage (V)
- Power directed towards the “load” (kW)
- Power directed towards the battery (kW)
- Energy produced daily (kWh)



On this screen you can see the current status of the battery pack. We may see four possible scenarios:

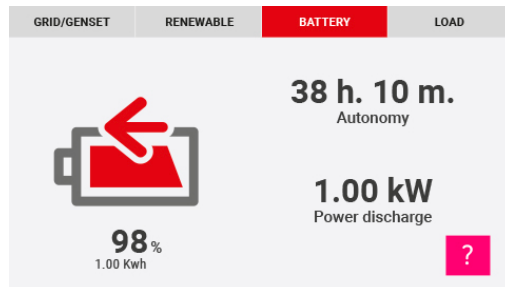
STATE OF CHARGE

On this screen you can see the current state of charge (SoC %), the power with which the battery block is being charged and the corresponding energy. The time remaining until the batteries are fully charged (at the current charging power) is also indicated.



You can view the battery status graphically and easily from the battery icon. Set out below are all the possible statuses of the battery while it is being charged.

STATE OF DISCHARGE

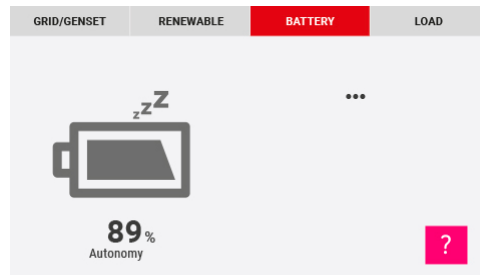


On this screen you can see the state of charge at the time of the consultation (SoC %), the power with which the battery block is being discharged and the corresponding energy. It also indicates the time remaining until the batteries are completely discharged or the autonomy time (at the current discharge power).



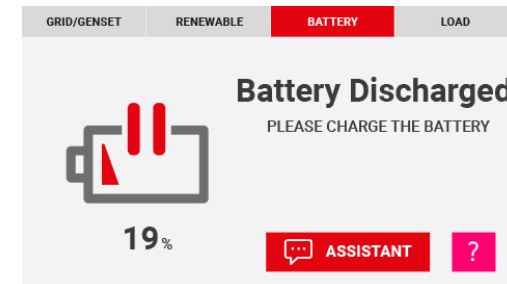
The discharge will be indicated by two battery icons that will warn the user of the approximate state of discharge.

IDLE STATUS



This screen indicates that the battery is idle, that is, it is neither charging nor discharging. This status can only appear when no operating mode is running or when the battery is fully charged, and the output is powered only by a generator set or an electrical distribution grid

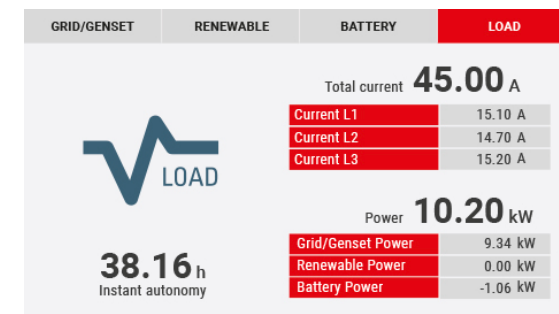
SHUTDOWN STATUS



This display indicates that the battery has a SoC of less than 20% and therefore the batteries must be charged. When the user observes this battery status, he or she can perform a charge immediately.

This screen shows the information generated by the electrical installation to be powered or "Charged". In this tab you can see the following data:

- Total current consumed (A)
- L1 phase current (A)
- L2 phase current (A)
- L3 phase current (A)
- Instantaneous power (kW)
- Primary power source power (kW)
- Battery power (kW)
- Solar panel power (kW)
- Battery life with current discharge power (h)



CONFIGURATION MENU

In the EHR window there is a button for setting and displaying a set of parameters. By pressing this button, the user will be able to access the “*CONFIGURATION*” tab.



On the top row of this screen, you can see selectable boxes. Using the navigation buttons you can access the two remaining tabs, “*CALENDAR*” and “*SETTINGS*”. Each of these configuration tabs is explained in more detail below.



The EHR allows you to see the parameters in detail and even to modify and configure other ones to ensure that the equipment meets the needs of the user correctly. To do this, by clicking on the configuration button you can see a list of readable parameters. These can be useful for the user, as all parameters are displayed without having to scroll through multiple screens to view the desired data. The data that we can see in this register are:

Date	Units	Block
Voltage phase 1	In	Input parameters
Voltage phase 2	In	Input parameters
Phase voltage 3	In	Input parameters
Current Phase 1	One	Input parameters
Current Phase 2	One	Input parameters
Current Phase 3	One	Input parameters
Frequency phase 1	Hz	Input parameters
Frequency phase 2	Hz	Input parameters
Frequency phase 3	Hz	Input parameters
Useful power phase 1	Kilowatt	Input parameters
Useful power phase 2	Kilowatt	Input parameters
Useful power phase 3	Kilowatt	Input parameters
DC Voltage	In	Battery parameters
Direct current	One	Battery parameters
DC Power	Kilowatt	Battery parameters
SOC	%	Battery parameters

State	-	Battery parameters
Voltage phase 1	In	Output parameters
Voltage phase 2	In	Output parameters
Phase voltage 3	In	Output parameters
Current Phase 1	One	Output parameters
Current Phase 2	One	Output parameters
Current Phase 3	One	Output parameters
Frequency phase 1	Hz	Output parameters
Frequency phase 2	Hz	Output parameters
Frequency phase 3	Hz	Output parameters
Useful power phase 1	Kilowatt	Output parameters
Useful power phase 2	Kilowatt	Output parameters
Useful power phase 3	Kilowatt	Output parameters
System Status	-	System parameters
Switch position	-	System parameters
Relay 1 CCGX	-	System parameters
Relay 2 CCGX	-	System parameters
DVCC System	One	System parameters
VE status. Bus	-	System parameters
VE error. Bus	-	System parameters
BMS Error VE. Bus	-	System parameters

CONFIGURATION	CUSTOMIZE CALENDAR		SETTINGS
PARAMETERS SYSTEM			
VE. Bus Reset			
READING PARAMETERS			
Input parameters		Output parameters	
Input voltage phase 1	23.00 V	Output voltage phase 1	23.00 V
Input voltage phase 2	23.00 V	Input voltage phase 2	23.00 V
Input voltage phase 3	23.00 V	Input voltage phase 3	23.00 V
Input frequency 1	23.00 V	Input frequency 1	23.00 V
Input frequency 2	23.00 V	Input power 1	23.00 kW
Input frequency 3	23.00 V	Input power 2	23.00 kW
Input power 1	23.00 kW	Input power 3	23.00 kW
Input power 2	23.00 kW	Battery parameters	
Input power 3	23.00 kW	Battery voltage	23.00 kW

You can also see a configuration element called “*VE. BUS RESET*”. This parameter is intended for resetting the EHR communications system. You will need to activate this parameter when the EHR behaves in an undesired way or when an alarm is triggered. This help function will indicate the activation of this parameter.

The EHR incorporates the possibility of configuring an operational calendar. This functionality is very useful if you want to restrict the operation of the machine when it is being used by third-party users. To configure the calendar, the user must indicate the days of the week, the time slot and the maximum power at which he or she wants to allow the use of the EHR

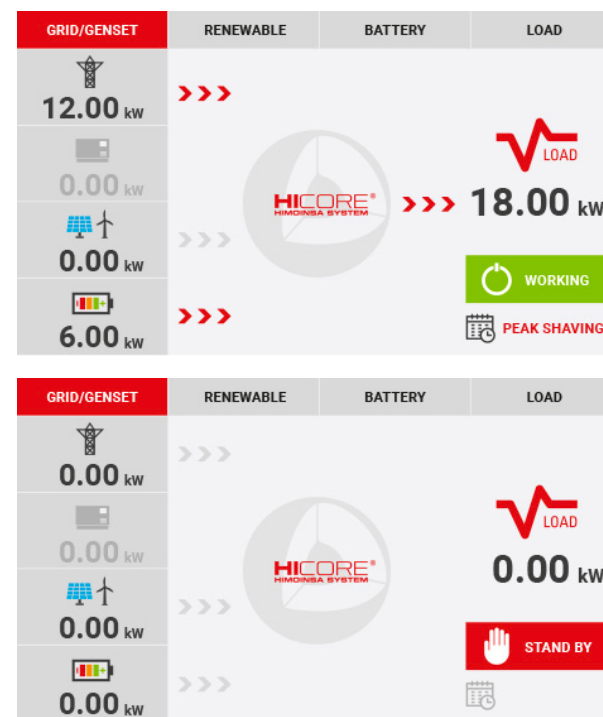
CONFIGURATION		CUSTOMIZE CALENDAR		SETTINGS

Once the calendar has been configured, the user can only use the machine in the configured time slot and when this slot is exceeded, a system shutdown will occur to indicate to the user that they have exceeded the enabled operating time. This will take place in the same way for the output power, since if the user exceeds the configured power, the system will shut down immediately, indicating to the user that they have exceeded the configured power.

If the calendar is active in the main menu, an image of a calendar will appear, thus indicating to the user that the use of the EHR is limited, and it may be the case that the machine cannot work because it is outside the configured time slot.

The “CALENDAR” icon will appear in light grey when the option is enabled and in

dark grey when we are within the time slot in which the EHR has been configured.



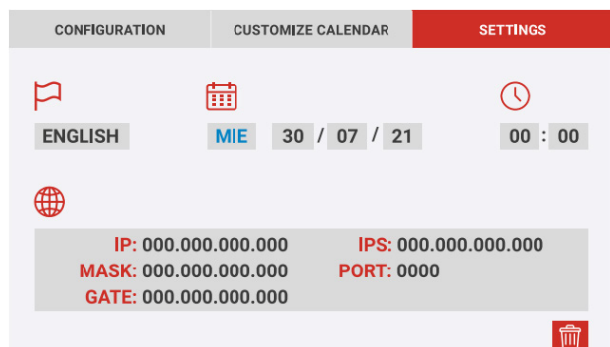
WARNING:

Remember to disable calendar ticks when you do not want to restrict the operation. Whenever one of these “ticks” is enabled , the calendar will be enabled, which will make it impossible for the equipment to operate outside the configured time slot.

WARNING:

Remember to set a high enough power to supply the desired installation as insufficient power would cause the system to shut down immediately.

In this screen we can modify internal parameters of the control panel, such as the time, date, language or the necessary settings for the grid generated by the EHR equipment



At the bottom of this tab, you can see a trash can icon. The user should use this when they want to clear all the configurations that have been made on the equipment and all the data generated. The user should note that as soon as this trash can icon is selected, the following data will be deleted:

- Configuration of the selected grid
- Selected language
- Selected day
- Selected time
- Selected calendar
- Alarm and warning logs
- Logs of consumed, generated and plotted power

The EHR equipment records all the data generated during its entire operation; this data can be displayed by clicking on the “*HISTORICAL DATA*” button.



On this screen you can see the historical data of different time periods, both in

graph and in data format. This is useful for tracking the consumption of different energy sources to monitor the efficiency of the system.



If the data display is selected by clicking on “*LIST*”, the following data layout appears.

ELAPS. T.	GRID/GENSET	RENEWABLE	B. CHARGE	B. DISCHARGE	LOAD
Now	0.00 kWh	0.00 kWh	0.00 kWh	0.00 kWh	0.00 kWh
1 Hour	123.81 kWh	150.35 kWh	123.81 kWh	150.35 kWh	23.81 kWh
2 Hours	22.18 kWh	49.61 kWh	22.18 kWh	49.61 kWh	22.18 kWh
3 Hours	6.62 kWh	1.03 kWh	6.62 kWh	1.03 kWh	6.62 kWh
4 Hours	7.16 kWh	0.00 kWh	7.16 kWh	0.00 kWh	7.16 kWh
5 Hours	123.81 kWh	50.35 kWh	123.81 kWh	50.35 kWh	23.81 kWh
6 Hours	7.71 kWh	249.61 kWh	7.71 kWh	249.61 kWh	7.71 kWh
7 Hours	6.62 kWh	1.03 kWh	6.62 kWh	1.03 kWh	6.62 kWh

When an alarm or warning occurs, the LED beside the “ALARMS AND WARNINGS” button will light up. When a warning or alarm occurs, the LED will flash but when the user enters the “ALARMS AND WARNINGS” screen, the LED will remain on all the time.



By pressing the button, you will be able to see what alarms and warnings are running at that moment. However, if we go to the alarm history, we can see up to the last 30 alarms and warnings that have occurred throughout the operation of the machine.

ELAPS. T.	GRID/GENSET	RENEWABLE	B. CHARGE	B. DISCHARGE	LOAD
Now	0.00 kWh	0.00 kWh	0.00 kWh	0.00 kWh	0.00 kWh
1 Hour	123.81 kWh	150.35 kWh	123.81 kWh	150.35 kWh	23.81 kWh
2 Hours	22.18 kWh	49.61 kWh	22.18 kWh	49.61 kWh	22.18 kWh
3 Hours	6.62 kWh	1.03 kWh	6.62 kWh	1.03 kWh	6.62 kWh
4 Hours	7.16 kWh	0.00 kWh	7.16 kWh	0.00 kWh	7.16 kWh
5 Hours	123.81 kWh	50.35 kWh	123.81 kWh	50.35 kWh	23.81 kWh
6 Hours	7.71 kWh	249.61 kWh	7.71 kWh	249.61 kWh	7.71 kWh
7 Hours	6.62 kWh	1.03 kWh	6.62 kWh	1.03 kWh	6.62 kWh

 GRAPHIC  LIST  00.00 kW  00.00 kWh

When you click on one of the alarms or warnings, a tab will appear with a help function to help the user proceed in the correct way. Nevertheless, for more information about the alarms and warnings produced by the system please refer to 7. *Warning and Alarms.*

6. MAINTENANCE

A proper maintenance and overhaul programme, performed only by qualified technicians, is essential if you wish to maximise inverter/battery reliability, minimise repairs and reduce long-term costs.

To carry out an effective maintenance programme, we recommend that you collect the data that are obtained during the work performed, using the EHR uptime meter to keep an accurate record of all the services performed. This record will also be important for security purposes.

To see the specific maintenance plans for the EHR you have purchased, please refer to the corresponding documentation. These plans vary depending on:

- Hours of operation
- Machine location
- Type of load it supplies
- Environmental conditions

In this way, once the equipment has been received and taking into account the aforementioned elements, these maintenance plans must be studied to determine how often maintenance needs to be carried out.

It is important to clean the EHR continuously and to also avoid the accumulation of liquids on both internal and external surfaces. Do not use flammable solvents when carrying out cleaning tasks; it is recommended that you use aqueous liquids for industrial cleaning.

IMPORTANT:

Before performing any operation, shut down the EHR and wait until it cools down.

The EHR requires a regular storage maintenance load when not in use. Connect it to an AC source all the time or at least every 4 weeks, ensuring that a full charge is maintained.

Battery maintenance charging must be done from a single-phase charging source, an electrical grid or a diesel generator set.

EHR equipment requires a regular maintenance load when not in use, to preserve its storage capacity. To do this, you should connect an AC source to the maintenance "SOCKET" all the time or at least once a week. In this way it is necessary to maintain a full load and to ensure that the equipment retains all its accumulation capacity.

The maintenance load must be carried out from a single-phase source, either a grid or a single-phase generator set. Remember that we can only supply batteries up to 16A AC through this input.

While the unit is charged, a constant single-phase AC supply must be connected to the maintenance input. The procedure for the maintenance load is as follows:

1. On the HICORE home screen press the "ASSISTANT" button
2. Specify that you want to perform a charge.
3. Specify that you want to perform a maintenance load
4. Connect the AC source to the mounting input connector on the maintenance panel. For more information about the location of this connector, see 5.6. EHR control panel.
5. Press "RUN"
6. The unit should now charge at low speed to prevent the battery from running out.

For more information, please refer to point 5.9. Battery charging.

6.1 BATTERY MAINTENANCE

While the unit is charged, a constant single-phase AC supply must be connected to the maintenance input. Follow this procedure for the maintenance load:

1. On the CORE home screen, press the "System Switch" button to turn the inverters off.
2. Turn the output switch off.
3. Connect the AC source to the mounting input connector on the maintenance control panel.
4. On the central home screen, press the "System Switch" button to turn the inverters on.

5. Turn the AC source switch on.

It is recommended that you leave the AC power supply constantly plugged in. If this is not possible, to maintain the product warranty it is necessary to charge the EHR unit at least every 4 weeks.

BATTERY DISCHARGES BELOW 10%

When the battery is discharged to 100% (0% SoC) follow these steps:

This procedure should only be performed by qualified personnel.

1. Turn the battery-inverter disconnect switch off; you will need a key for this.
2. Push the continuous disconnect (DC) switch down.
3. Open the lower side panels to access the batteries.
4. Click the ON switch on each of the batteries and turn all the batteries in the unit on.
5. Press the SW button on battery No. 1 and wait for all batteries to start.
6. Connect the external DC power supply to the DC distribution bar
7. Set the power supply to 55 V and let the batteries charge until the voltage reaches 51 V.
8. If there is a module with the red ERR LED flashing, that module or modules should be charged individually following this procedure:
 - a. Turn all the modules off by pressing the red button on battery number one for a few seconds (this battery will be indicated); then press each individual kill switch.
 - b. Disconnect the power cords from the alarmed module. Isolate the terminals from the wires to avoid short circuits.
 - c. Disconnect the communication cables.
 - d. Connect the DC power supply to the DC+ and DC- terminals
 - e. Set the power supply to 55 V and charge the modules until the voltage reaches 51 V.

f. Reconnect the battery modules to the system by reconnecting the power cords and communication cables.

g. Click the ON button. All the modules should be activated.

1. All system modules should have the SoC indicator LEDs on and flashing in green without any ERROR or ALARM LEDs.
2. Disconnect the DC power supply.
3. Wait until the BMU LED stops flashing and remains solid green. This may take several minutes.
4. Check that the LEDs on each of the battery modules are now also solid green and are not flashing.
5. Press the red button on battery number one (it will be indicated) for 2s. This will turn the batteries off again.
6. Battery-inverter insulators (rotary switches located under the inverters) can be set to "ON" and then you have to wait 20 seconds; the key is needed to do this.
7. Press the red button on battery number one for several seconds (it will be indicated).
8. Wait about 2 minutes.
9. When you see that all the battery status LEDs are green, we can continue.
10. Turn on the automatic continuous power (DC) switch intended to power the system control.
11. Connect a power supply to the AC-IN input to the EHR.
12. Perform a maintenance load, see Battery Charging section.
13. The EHR should begin charging the batteries. Allow the unit to reach a charge of 100%.
14. Once the battery's SoC level has reached 100%, close the side door panels.
15. The unit is now initialised. Refer to the user manual for setup and operating instructions.

ALARM IN A BATTERY STRING MODULE

If the operation of one of the battery modules encounters an alarm (indicated by a flashing red LED), it is best to disconnect this battery from the system and continue working with the rest of the batteries (remember that this action causes

a decrease in the autonomy of the batteries). The procedure for disconnection due to the failure of a battery module is as follows:

This procedure should only be performed by qualified personnel.

16. See if the control panel is displaying an alarm or a warning related to the battery error, see 7. Warning and Alarms.
17. Disconnect the HDI by following the procedure in section EHR shutdown procedure.
18. Disconnect the power cords from the faulty battery bank module. To proceed with the disconnection, remove the battery's P+ and P- power terminals. To do this, click on the push button located on each of the connectors and remove it from the connection terminal.
19. Make sure the power connectors are properly insulated and secure once removed, as there is a high potential risk of short circuiting if either of these terminals touch any part of the unit.
20. Connect the DC power supply to the DC+ and DC terminals.
21. Set the power supply to 55 V and charge the modules until the voltage reaches 51 V.
22. If it is properly charged, it should be returned to the battery bank; to do this, please follow the procedure set out in EHR Shutdown procedure.
23. If the battery has not been charged properly, the battery should be replaced with a new one (make sure the same battery model is used).

BATTERY BALANCING.

If a battery module is changed, the module must be balanced before it is integrated into the EHR system. To do this, please follow these steps carefully.

This procedure should only be performed by qualified personnel.

1. Leave the EHR equipment in "STAND BY" (no operating or charging mode is running)
2. Measure the voltage on the DC busbar.
3. Measure the voltage between the terminals of the new battery module
4. If the voltage of the busbar is higher than the voltage of the new module, the module must be charged externally up to the voltage measured in the busbar DC.
5. If the voltage measured in the DC busbar is lower than that measured in the new module, the EHR batteries must be discharged to the voltage measured in the new battery module. For this it is advisable to use the "LOW LOADS" mode with a small discharge power ([1 – 2] kW), see 3.2.5. LOW LOADS mode.
6. Once the voltage is equalised, the new module can be connected to the other batteries. To do this, the shutdown procedure must be carried out beforehand, see EHR shutdown procedure section.
7. The module can then be inserted and the P+ and P- power cables and then the communications cables can be connected.
8. After the connection is made, you can continue with the start-up procedure, see EHR start-up procedure.
9. The unit is now initialised. Refer to the user manual for configuration and operating instructions.

STORAGE MAINTENANCE

The procedure for performing rotational storage maintenance is:

1. Perform a battery maintenance charge until a 100% state of charge is reached.
2. To maintain the product warranty it is necessary to perform a maintenance load of the EHR unit at least every 4 weeks.

6.2 RECOMMENDED INSPECTION OF RENTAL EQUIPMENT

Checks should be carried out on returned units to ensure that the EHR is in good condition.

Subsequent inspection:

- All housings are free of dust, dirt or scratches.
- All necessary screws are in place and tightened tightly.
- The lifting terminal is tightened to the correct torque (250 Nm).
- Keys are in place.
- All doors except the upper front one are locked.
- The battery bank has a SoC $\geq 99\%$
- The INVERTER DC disconnectors are in the right position.
- The unit can be turned off and on by selecting operating modes on the HICORE. Leave the unit off.
- The control panel and EHR are properly connected and the SIM card contains data.
- Check any alarms produced during the rental period. See 7. Warning and Alarms.
- Update the HICORE time if necessary.
- Check the air filters and replace them if they are dirty.
- Return the unit to the default settings via HICORE. This action will delete any configurations that have been made and any alarms and logs that have been produced.

If all of the above points are satisfied, the unit will be ready for storage, see 4. Unloading, handling and transport. The default settings can be restored via HICORE as follows:

1. Go to the " Configuration" screen
2. Navigate to the " Configuration" indicator
3. Navigate to the "Trash can" icon
4. Click on "OK"

6.3 DISCONNECTION OF STORAGE EQUIPMENT

FOR THE INSTALLER

The EHR should always be kept on and charged. However, if the unit is to be shipped, transported, or cannot be accessed due to maintenance for more than 4 weeks, the batteries must be turned off to avoid excessive discharge. The following steps must be followed in the correct order:

1. Make sure the EHR is 100% charged.
2. Finish the charging or mode of operation that is being performed. Make sure that the HICORE home screen shows "STAND BY".
3. Use the start-up button to turn the control panel off (press for 5 seconds).
4. Disconnect all system inputs and outputs.
5. Open the doors of the EHR and you will see a switch on the panel of each of the inverters. This switch must be in the "OFF" position (at the switching midpoint).

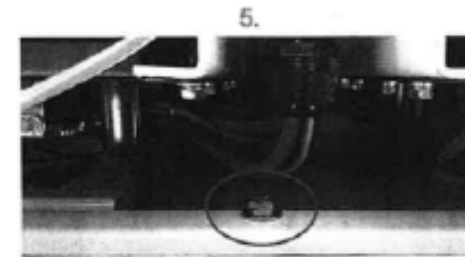


6. Push the continuous disconnecter (DC) switch down

7. Turn the disconnectors between each of the inverters and the battery off (they are located beneath the inverters on the motherboard). Please note that depending on the configuration of the unit, the insulators can be accessed from either side door.



8. Remove the lower side panels that cover the battery sections on either side of the unit by unscrewing the 3 M6 hexagonal head bolts hidden inside the unit.



9. Set the on/off power switch on each of the batteries to off. This switch is integrated into all battery modules so that each battery module can be turned off.
10. Make sure all BMU battery modules, DC-DC converter, modem, and C2CLUD are turned off. No LEDs should be flashing inside the unit.
11. Verify that there is no voltage on the DC busbar.
12. Replace the lower side panels of the battery and close the upper side doors
13. The unit is now completely disabled.

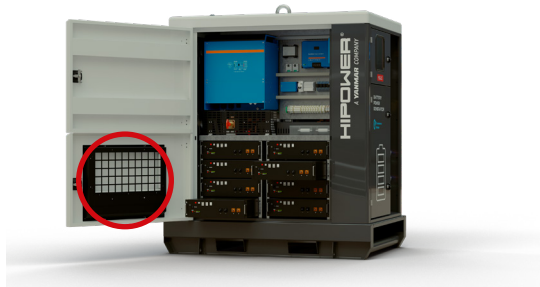
Follow the EHR start-up procedure when you turn the unit back on.

6.4 FILTER MAINTENANCE

There are two filters in your EHR. One is located on the back of the EHR just before the **HCVA G3** ventilation fans (see image below):



And the second filter, a type G4 panel filter, is located in the air inputs on both sides (image below).



Please check the condition of the filter regularly, according to the operating conditions of the unit

7. WARNING AND ALARMS

This section describes the alarms and warnings that may occur during the operation of the EHR equipment. These alarms must be heeded and as far as possible resolved, which is why a comment is attached next to each alarm and warning to guide the user when he or she has to resolve said alarm or warning. Whenever an alarm occurs, the system will shut down the operating mode and until this alarm is dealt with and resolved it will not be able to use the EHR equipment again. To differentiate between an alarm and a warning, an on-screen abbreviation is used, thus:

Subject	Abbreviation
Warning	E
Alarm	S

To solve the alarms it is of vital importance to attend to the block in which they are generated. We can distinguish between 4 fundamental blocks:

Subject	Abbreviation
Inverters	Y
Battery	B
Renewable	R
Central	C

This classification gives us an approximation of where the problem occurs and how to deal with it. Certain alarms and warnings appear more readily than others and these are explained below in more detail.

We label these warnings and alarms as frequent, but most of them appear when the equipment is under stress and we are approaching its maximum capabilities. Below is a small table that lists the alarms and warnings that occur most frequently and then explains why they occur.

WARNINGS

INDICATOR	BLOCK	SUBJECT
Warning #1 High inverter temperature	Inverters	Warning
Warning #2 Low battery	Inverters	Warning
Warning #3 Output overload	Inverters	Warning
Warning #7 Incorrect phase rotation	Inverters	Warning
Warning #10 The state of charge of the battery is 30%	Central	Warning
Warning #11 The state of charge of the battery is 25%	Central	Warning
Warning #12 The state of charge of the battery is 21%	Central	Warning
Incorrect phase rotation	Inverters	Warning
Warning #10 The state of charge of the battery is 30%	Central	Warning
Warning #11 The state of charge of the battery is 25%	Central	Warning
Warning #12 The state of charge of the battery is 21%	Central	Warning

ALARMS

INDICATOR	BLOCK	SUBJECT
Alarm #1 The equipment has shut down because one of the phases has been disabled	Inverters	Alarm
Alarm #2 Not all expected devices are in the system	Inverters	Alarm
Alarm #4 A time synchronisation error has occurred	Inverters	Alarm
Warning #11 The state of charge of the battery is 25%	Central	Warning
Alarm #13 Output overload	Inverters	Alarm
Alarm #38 Low battery voltage	Battery	Alarm
Alarm #39 High battery voltage	Battery	Alarm
Alarm #51 Emergency mushroom	Central	Alarm
Alarm #52 Differential switch jump	Central	Alarm
Alarm #53 Low battery operating mode shutdown	Central	Alarm

INDICATOR	BLOCK	SUBJECT
Alarm #55 The maximum contracted power has been exceeded	Central	Alarm
Low battery voltage	Battery	Alarm
Alarm #39 High battery voltage	Battery	Alarm
Alarm #51 Emergency mushroom	Central	Alarm
Alarm #52 Differential switch jump	Central	Alarm
Alarm #53 Low battery operating mode shutdown	Central	Alarm
Alarm #55 The maximum contracted power has been exceeded	Central	Alarm

Keep in mind that these warnings and alarms are not the only ones in the system. The list of these can be seen in section 7. Warning and Alarms.

These warnings and alarms will be described below. We will list the warnings first.

HIGH INVERTER TEMPERATURE (WARNING #1)

There are two possible causes:

1. Very high ambient temperature: Therefore, the inverters are cooled down at too hot a temperature, thereby causing a decrease in performance.
2. Very high inverter power: This causes the inverters to heat up internally, since although the ventilation is adequate, with an ambient temperature above 30°C it is very likely that this warning will appear. It should be noted that this only happens when we approach the inverter limits.

The activation of this warning falls on the temperature wing. With more or less time depending on the power that is being inverted and the ambient temperature at that time.

It is advisable to deal with this warning when it appears in order to prevent the temperature alarm from occurring as per the help functions shown on the screen. These can also be seen in 7. Warning and alarms.

LOW BATTERY (WARNING #2)

This warning occurs when the battery voltage begins to decrease below a given threshold; this warning does not give rise to any problems as when the SoC is less than 20% a shutdown occurs, and the safety margins are therefore adequate.

If this warning does appear, it is telling the user that the battery is running low. For more information, see 7. Warning and alarms.

OUTPUT OVERLOAD (WARNING #3)

This warning occurs when a high inversion is being made, close to the limits presented by the EHR. It is directly related to the temperature warning, since when an overload warning occurs and the ambient temperature is high, the system tends to increase its internal temperature and produces a temperature warning. If it occurs the other way round, the behaviour is similar: when a temperature warning occurs the system tends to worsen the performance and the EHR increases the inversion power causing an overload warning. For more information on nominal inversion power values as a function of ambient temperature, see 3.3. Technical characteristics.

The activation of this warning stems from the overload alarm. With more or less time depending on the power that is being inverted and the ambient temperature at that moment.

It is advisable to deal with this warning when it appears in order to prevent the temperature alarm from occurring as per the help functions shown on the screen. These can also be seen in 7. Warning and alarms.

INCORRECT PHASE ROTATION (WARNING #7)

This warning appears when the EHR equipment tries to synchronise with the grid and cannot do so because the phase sequence is invalid, that is, it informs the user that the connected phase sequence (either busbar or "SOCKET") is not suitable (it performs the function of a phasemeter). When this warning appears, the user must shut the EHR equipment down, reverse the sequence of a couple of phases and then reassemble the system and the EHR should be ready to synchronise with the grid.

For more information, please refer to the on-screen help or see 7. Warning and alarms.

LOW SOC STATUS (WARNING #10, WARNING #11 AND WARNING #12)

This warning appears when the charge is only powered by batteries and the state

of charge presented by the batteries (SoC) is low. Prompt loading or maintenance is recommended when this warning occurs.

Depending on the operating mode that is currently running, the battery charge will run automatically or not when the SoC is less than 20%. For more information, see 5.10 Working modes.

THE EQUIPMENT HAS BEEN TURNED OFF BECAUSE ONE OF THE PHASES HAS BEEN TURNED OFF (ALARM #1)

This alarm occurs for several reasons, as it indicates that the master phase has either been deactivated for system security reasons or due to repeated shutdown of the equipment. When the emergency mushroom is pressed, this alarm appears with the "Emergency Mushroom" alarm, since the emergency mushroom causes a shutdown of the master phase of the equipment and performs an immediate system shutdown. Another cause can be when there is an unsafe condition for the equipment, for example when there is a very high overload, and the machine shuts down suddenly; in most cases the alarm will appear due to the shutdown and this alarm.

If this alarm is caused by pressing on the mushroom, disengaging it will suffice, but if this alarm occurs because of any other reason and continues to run continuously, the system should be reset. For more information, see 5.10 Working modes.

NOT ALL EXPECTED ELEMENTS HAVE BEEN FOUND IN THE SYSTEM (ALARM #2)

This alarm occurs when the system loses communication with one of the inverters, either L1, L2 or L3. Communication with the inverter can fail for several reasons, including:

- False contacts in the DC distribution of the inverter in question, either due to the "battery switch" or incorrect tightening torques on the terminals.
- Poor connection of the communication cables between the inverters and the L1 inverter to the Cerbo GX, due to a worn RJ45 connector or a broken.
- One of the inverters is defective and has stopped working due to internal causes.

This alarm can be related to the previous alarm, "The equipment has been turned off because one of the phases has been turned off" (Alarm #1), since, if an immediate shutdown occurs due to one of the inverters being turned off during its operation, both alarms can occur instantly. To solve the execution of this alarm, you should refer to the on-screen help or see 7.Warning and Alarms.

A TIME SYNCHRONISATION ERROR HAS OCCURRED (ALARM #4)

This alarm appears when the synchronisation between the inverters is erroneous, due to a failure of real-time communications. As a result, the system issues an alarm and shuts the operating mode down. In the vast majority of cases this alarm runs continuously and it is necessary to perform a complete restart of the EHR equipment to resolve it, see 5. Starting up the EHR plant. Although it is advisable to perform the steps indicated in the on-screen help, see 7.Warning and Alarms.

HIGH INVERTER TEMPERATURE (ALARM #11)

This alarm occurs after the "High inverter temperature" warning. For this reason, you have to previously deal with the warning by referring to the on-screen help, please also refer to 7.Warning and Alarms.

OUTPUT OVERLOAD (ALARM #13)

This alarm occurs after the "Output overload" warning. For this reason, you have to previously deal with the warning by referring to the on-screen help, please also refer to 7.Warning and Alarms.

LOW VOLTAGE BATTERY (ALARM #38)

This alarm occurs because the BMS of the batteries detects that a cell of the batteries has a voltage that is lower than acceptable, for this reason the system shuts the operating mode down. It is worth mentioning that the best way to solve this problem is to charge the battery, either a fast or a maintenance charge (if a maintenance charge is performed, the internal cells of the batteries should be balanced to achieve a better result). For more information, refer to the on-screen help or to 7.Warning and Alarms.

HIGH VOLTAGE BATTERY (ALARM #39)

This alarm occurs because the BMS of the batteries detects that a cell of the batteries has a voltage that is higher than acceptable, for this reason the system shuts the operating mode down. It is worth mentioning that the best way to solve

this problem is to discharge the batteries a little until the alarm is resolved (Low charges, for example) and then perform a maintenance charge so that the internal cells of the batteries are balanced. For more information, refer to the on-screen help or to 7.Warning and Alarms.

EMERGENCY MUSHROOM (ALARM #51)

This alarm occurs due to the pressing of the emergency mushroom. To eliminate this alarm the mushroom must be rearmed. Please note that if the emergency mushroom has been pressed, it may be due to a safety condition. Check the condition of the machine and check with the operator who has pressed the emergency mushroom before rearming it. For more information, refer to the on-screen help or to 7.Warning and Alarms.

DIFFERENTIAL JUMP (ALARM #52)

This alarm is due to the differential switch of the EHR equipment that is running. This is due to a current drift at the output. Before reassembling the magnetothermal output check all the wiring and the tightening torque of the terminals. If all is correct, it is possible that this drift is occurring in the installation that is feeding the EHR, so you should check that installation. For more information, refer to the on-screen help or to 7.Warning and Alarms.

Once the problem is solved, reset the magnetothermal output and then rearm the differential switch.

LOW BATTERY OPERATING MODE STOP (ALARM #53)

This alarm occurs when any operating mode is working and the state of charge of the batteries or SoC has decreased by 20%: for safety reasons, the operating mode is shut down. This alarm occurs when the primary power supply cannot be activated remotely or it is supplying the installation at its maximum power, making it impossible to charge the batteries. It is advisable to charge the battery when this alarm occurs. For more information, refer to the on-screen help or to 7.Warning and Alarms.

THE MAXIMUM CONTRACTED POWER HAS BEEN EXCEEDED (ALARM #55)

This alarm occurs when the calendar is enabled. When the calendar is configured, a period of time and a maximum power will be established. When the installation is supplied with a power higher than that which has been configured, the mode of operation at that moment will be shut down immediately. For more information about configuring the calendar, see Calendar section.

You may wish to refer to the on-screen help for more information, or to 7.Warning and Alarms.

The following is a list of all warnings and alarms that may occur in the system along with their respective help functions to solve each of them.

7.1 WARNINGS

INDICATOR	ORIGIN	TYPE
Warning #1 High inverter temperature	Inverters	Warning
Warning #2 Low battery	Inverters	Warning
Warning #3 Output overload	Inverters	Warning
Warning #4 Ripple L1	Inverters	Warning
Warning #5 Ripple L2	Inverters	Warning
Warning #6 Ripple L3	Inverters	Warning
Warning #7 Incorrect phase rotation	Inverters	Warning
Warning #8 Warning on one of the battery slaves	Battery	Warning
Warning #9 Low cell voltage	Battery	Warning
Warning #10 The state of charge of the battery is 30%	Central	Warning
Warning #11 The state of charge of the battery is 25%	Central	Warning
Warning #12 The state of charge of the battery is 21%	Central	Warning
Warning #7 Incorrect phase rotation	Inverters	Warning
Warning #8 Warning on one of the battery slaves	Battery	Warning
Warning #9		

Low cell voltage	Battery	Warning
Warning #10		
The state of charge of the battery is 30%	Central	Warning
Warning #11		
The state of charge of the battery is 25%	Central	Warning
Warning #12		
The state of charge of the battery is 21%	Central	Warning

WARNINGS HELP

INDICATOR	HELP
Warning #1	Disconnect the load at the power output. If the warning persists, disconnect the primary power supply.
Warning #2	Charge the battery. You can charge the batteries by applying an operating mode and performing a maintenance charge. If this is not possible, refer to the manual to see how to charge the batteries externally.
Warning #3	Decrease the power consumption. If this is not possible, disconnect the load at the output.
Warning #4	Possible internal battery problem: 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the batteries are working properly. Check to see if the error LEDs are on (red). 3. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 4. Check that the voltage between the DC terminals at the input of each of the inverters is correct. 5. Turn the batteries off and restart the equipment. 6. If the error persists, it will be because one or more batteries are defective.
Warning #5	Possible internal problem of lotteries: 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the batteries are working properly. Check to see if the error LEDs are on (red). 3. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 4. Check that the voltage between the DC terminals of the input of each of the inverters is correct. 5. Turn the batteries off and restart the equipment. 6. If the error persists, it will be because one or more batteries are defective.

INDICATOR	HELP
Warning #6	Possible internal battery problem: 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the batteries are working properly. Check to see if the error LEDs are on (red). 3. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 4. Check that the voltage between the DC terminals of the input of each of the inverters is correct. 5. Turn the batteries off and restart the equipment. 6. If the error persists, it will be because one or more batteries are defective.
Warning #7	The arrangement of the phases is not correct. Redistribute them.
Warning #8	Possible internal battery problem: 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 3. Check that the RJ45 connectors are in good condition and tight. 4. Check that the batteries are working properly. Check to see if the error LEDs are on (red). If they are, identify which battery is defective. 5. Turn the batteries off and restart the equipment. 6. If the error persists, it will be because one or more batteries are defective. It can be both a deterioration of the cells and an internal communication failure.
Warning #9	Possible internal battery problem: 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 3. Check that the RJ45 connectors are in good condition and tight. 4. Check that the batteries are working properly. Check to see if the error LEDs are on (red). If they are, identify which battery is defective. 5. Turn the batteries off and restart the equipment. 6. If the error persists, it will be due to a possible problem in one or more battery modules, either due to a deterioration of the cells or due to a problem in communication.
Warning #10	The state of charge of the battery is 30%. Once it reaches 20%, the operating mode will be shut down. Please charge the battery as soon as possible.
Warning #11	The state of charge of the battery is 25%. Once it reaches 20%, the operating mode will be shut down. Please charge the battery as soon as possible.
Warning #12	The state of charge of the battery is 21%. Once it reaches 20%, the operating mode will be shut down. Please charge the battery as soon as possible.

7.2 SYSTEM ALARMS

INDICATOR	ORIGIN	TYPE
Alarm #1 The equipment has shut down because one of the phases has been disabled	Inverters	Alarm
Alarm #2 Not all expected devices are in the system	Inverters	Alarm
Alarm #3 Surge in ACOUT	Inverters	Alarm
Alarm #4 A time synchronisation error has occurred	Inverters	Alarm
Alarm #5 Device cannot transmit data	Inverters	Alarm
Alarm #6 One of the devices assumed the master state because the original master failed	Inverters	Alarm
Alarm #7 An AC overload has occurred at a slave output while it is turned off	Inverters	Alarm
Alarm #8 This device cannot function as a slave	Inverters	Alarm
Alarm #9 Firmware incompatibility. The firmware of one of the connected devices is not up-to-date enough to work with this device	Inverters	Alarm
Alarm #10 Internal error in inverters	Inverters	Alarm
Alarm #11 High inverter temperature	Inverters	Alarm
Alarm #12 Low battery	Inverters	Alarm
Alarm #13 Output overload	Inverters	Alarm
Alarm #14 Ripple L1	Inverters	Alarm
Alarm #15 Ripple L2	Inverters	Alarm
Alarm #16 Ripple L3	Inverters	Alarm
Alarm #17 BMS Error	Inverters	Alarm
Alarm #18 Error in the MPPT	MPPT	Alarm
Alarm #19 Battery initialisation failed	Battery	Alarm
Alarm #20 No batteries connected	Battery	Alarm

INDICATOR	ORIGIN	TYPE
Alarm #21 Connected battery is not recognised	Battery	Alarm
Alarm #22 Battery initialisation	Battery	Alarm
Alarm #23 Incorrect battery initialisation	Battery	Alarm
Alarm #24 Battery measurement error	Battery	Alarm
Alarm #25 Error in the internal calculation of the batteries by the BMS	Battery	Alarm
Alarm #26 Battery blocking error	Battery	Alarm
Alarm #27 Shut down standby batteries	Battery	Alarm
Alarm #28 Error with pre-load power	Battery	Alarm
Alarm #29 An error occurred in the battery safety contactor	Battery	Alarm
Alarm #30 Error with pre-load discharge	Battery	Alarm
Alarm #31 Battery ADC error	Battery	Alarm
Alarm #32 Error in one of the battery cells	Battery	Alarm
Alarm #33 Battery pre-load error	Battery	Alarm
Alarm #34 Battery safety contactor error	Battery	Alarm
Alarm #35 Update error on one of the battery slaves	Battery	Alarm
Alarm #36 Invalid battery slave update	Battery	Alarm
Alarm #37 General battery alarm	Battery	Alarm
Alarm #38 Low battery voltage	Battery	Alarm
Alarm #39 High battery voltage	Battery	Alarm
Alarm #40 High battery temperature	Battery	Alarm
Alarm #41 Low battery temperature	Battery	Alarm
Alarm #42 High charging current	Battery	Alarm
Alarm #43 High discharge current	Battery	Alarm
Alarm #44 High internal battery temperature	Battery	Alarm

INDICATOR	ORIGIN	TYPE
Alarm #45 Internal battery failure	Battery	Alarm
Alarm #46 Low state of charge	Battery	Alarm
Alarm #47 Cell imbalance	Battery	Alarm
Alarm #48 Low cell voltage	Battery	Alarm
Alarm #49 High charging temperature	Battery	Alarm
Alarm #50 Low charging temperature	Battery	Alarm
Alarm #51 Emergency mushroom	Central	Alarm
Alarm #52 Differential switch jump	Central	Alarm
Alarm #53 Low battery operating mode shutdown	Central	Alarm
Alarm #54 Shutdown of operating mode due to communication failure	Central	Alarm
Alarm #55 The maximum contracted power has been exceeded	Central	Alarm

7.3 HICORE ALARMS

INDICATOR	HELP
Alarm #1	An error occurred in one of the inverters either because the maintenance disconnecter is damaged or the inverter itself has been damaged. 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the communication between the inverters is correct. 3. Check that the RJ45 connectors are in good condition and tight. 4. Check the position of the inverter disconnecter causing the error. 5. To ensure the disconnecter is in the correct position, turn the batteries off and replace the disconnecter. 6. Restart the equipment. 7. If the error persists, it is possible that one of the inverters has stopped working.

INDICATOR	HELP
Alarm #2	An error occurred in one of the inverters. 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the communication between the inverters is correct. 4. Check that the RJ45 connectors are in good condition and tight. 5. Check that the voltage between the DC terminals of the input of each of the inverters is correct. 6. If not, the maintenance disconnecter may have broken. This element will need to be replaced. 7. Turn the batteries off and restart the equipment. 8. If the error persists, it is possible that one of the inverters has stopped working.
Alarm #3	An error has occurred in the inverters due to a problem in the installation or an internal problem in the inverters themselves. 1. Disconnect the load at the power output. 2. Run any operating mode. 3. Verify by means of a voltmeter that the RMS voltage at the output is higher than adequate. 4. If it is, one of the inverters may have stopped working. 5. If it is not, connect the load at the output and verify that there is in fact an overvoltage. 6. If there is, the problem has to do with the electrical installation carried out on the load.
Alarm #4	An error has occurred in the inverters. 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the communication between the inverters is correct. 3. Check that the RJ45 connectors of the inverters and the Cerbo GX are in good condition and tight. 4. Turn the batteries off and restart the equipment.
Alarm #5	System communication failed. 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the communication between the inverters and between the Cerbo GX and the master inverter is correct. 3. Turn the batteries off and restart the equipment. 4. If the problem persists, a problem has occurred in the Cerbo GX or in one of the inverters.
Alarm #6	An error occurred in one of the inverters. 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the communication between the inverters is correct, especially the L1 inverter. 4. Check that the RJ45 connectors of the L1 inverter are in good condition and tight. 5. Check that the voltage between the DC terminals of the L1 inverter input is correct. 6. If not, the maintenance disconnecter may have broken. This element will have to be replaced. 7. Turn the batteries off and restart the equipment. 8. If the error persists, it is possible that the L1 inverter has stopped working.

INDICATOR	HELP
Alarm #7	A problem has occurred in the installation connected to the load. 1. Disconnect the currently connected load. 2. Connect another load different from the one connected earlier. 3. Verify that this error is not running again. 4. Check the electrical installation connected to the load.
Alarm #8	An error occurred in the communication of one of the inverters. 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the communication between the inverters is correct. 4. Check that the RJ45 connectors of the inverters and the Cerbo GX are in good condition and tight. 5. Turn the batteries off and restart the equipment. 6. If the problem persists it is possible that one of the inverters has stopped working.
Alarm #9	An error has occurred in the firmware. 1. Contact technical support for a firmware update.
Alarm #10	An error occurred in one of the inverters. 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the communication between the inverters is correct. 4. Check that the RJ45 connectors of the inverters and the Cerbo GX are in good condition and tight. 5. Check that the voltage between the DC terminals of the input of each of the inverters is correct. 6. The maintenance disconnecter may have broken. This element will need to be replaced. 7. Turn the batteries off and restart the equipment. 8. If the error persists, it is possible that one of the inverters has stopped working.
Alarm #11	Disconnect the load at the power output. If it persists, disconnect the primary power supply.
Alarm #12	Charge the battery. You can charge the batteries by applying an operating mode and performing a maintenance charge. If this is not possible, please refer to the manual to see how to charge the batteries externally.
Alarm #13	Decrease the power consumption. If this is not possible, disconnect the load.
Alarm #14	Possible internal battery problem: 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the batteries are working properly. Check to see if the error LEDs are on (red). 3. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 4. Check that the voltage between the DC terminals of the L1 inverter input is correct. 5. Turn the batteries off and restart the equipment. 6. If the error persists, it will be because one or more batteries are defective.

INDICATOR	HELP
Alarm #15	Possible internal battery problem: 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the batteries are working properly. Check to see if the error LEDs are on (red). 3. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 4. Check that the voltage between the DC terminals of the L2 inverter input is correct. 5. Turn the batteries off and restart the equipment. 6. If the error persists, it will be because one or more batteries are defective.
Alarm #16	Possible internal battery problem: 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the batteries are working properly. Check to see if the error LEDs are on (red). 3. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 4. Check that the voltage between the DC terminals of the L3 inverter input is correct. 5. Turn the batteries off and restart the equipment. 6. If the error persists, it will be because one or more batteries are defective.
Alarm #17	An error has occurred in the BMS of the batteries. 1. Refer to the manual to perform a system shutdown (do not turn the batteries off) 2. Check that the batteries are working properly. Check to see if the error LEDs are on (red). 3. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 4. Turn the batteries off and restart the equipment. 5. If the error persists, it will be because one or more batteries are defective.
Alarm #18	The MPPT is not working properly or is not operational. 1. Disconnect the solar input applied to the EHR 2. Check that the MPPT is in good visual condition (not swollen or cracked). 3. Check that the communication between the Cerbo GX and the MPPT is correct. This is a special connector called "VE. Direct." 3. Apply a DC supply to the solar input terminals (>55V and >10A). 4. Verify that the magneto-technical protections are enabled. 5. Verify that the fuses are operational. 6. If the error persists, it will be necessary to replace the MPPT equipment.
Alarm #19	Possible internal battery problem: 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the batteries are working properly. Check to see if the error LEDs are on (red). 3. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 4. Check that the voltage between the DC terminals of the input of each of the inverters is correct. 5. Turn the batteries off and restart the equipment. 6. If the error persists, it will be because one or more batteries are defective.

INDICATOR	HELP
Alarm #20	Possible internal battery problem: 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the batteries are working properly. Check to see if the error LEDs are on (red). If they are, identify the defective elements. 3. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 4. Check that the RJ45 connectors connected between the battery modules are in good condition and tightly tightened. 5. Check that the voltage between the DC terminals of the input of each of the inverters is correct. 6. Turn the batteries off and restart the equipment. 7. If the error persists, it will be because one or more batteries are defective.
Alarm #21	Possible internal battery problem: 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the batteries are working properly. Check to see if the error LEDs are on (red). If they are, identify the defective elements. 3. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 4. Check that the RJ45 connectors connected between the battery modules are in good condition and tightly tightened. 5. Check that the voltage between the DC terminals of the input of each of the inverters is correct. 6. Turn the batteries off and restart the equipment. 7. If the error persists, it will be because one or more batteries are defective.
Alarm #22	Possible internal battery test problem: 1. Click on the variable "VE. Reset" to reset the communication. For more information, refer to the manual. 2. If the alarm continues to occur, refer to the manual to perform a system shutdown (do not turn the batteries off). 3. Check that the batteries are working properly. Check to see if the error LEDs are on (red). If they are, identify the defective elements. 4. Check that the RJ45 connectors connected between the battery modules are in good condition and tightly tightened. 5. Check that the voltage between the DC terminals of the input of each of the inverters is correct. It must have a value between [45 – 52] V. 6. Turn the batteries off and restart the equipment. 7. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. To do this, you can observe the parameters of the battery and check that they are logical. 8. If the error persists, it will be because one or more batteries are defective.
Alarm #23	Possible internal battery problem: 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the batteries are working properly. Check to see if the error LEDs are on (red). If they are, identify the defective elements. 3. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 4. Check that the RJ45 connectors connected between the battery modules are in good condition. 5. Check that the voltage between the DC terminals of the input of each of the inverters is correct. 6. Turn the batteries off and restart the equipment. 7. If the error persists, it will be because one or more batteries are defective.

INDICATOR	HELP
Alarm #24	Possible problem in the BMS of the batteries. 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the batteries are working properly. Check to see if the error LEDs are on (red). If they are, identify the defective elements. 3. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 4. Check that the RJ45 connectors connected between the battery modules are in good condition and tightly tightened. 5. Check that the voltage between the DC terminals of the input of each of the inverters is sufficiently close to the voltage that appears on the battery screen. 6. Turn the batteries off and restart the equipment. 7. If the error persists, it will be due to the fact that the voltage pressure of the BMS of the batteries is not working properly. This module must be replaced.
Alarm #25	Possible problem in the BMS of the batteries. 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the batteries are working properly. Check to see if the error LEDs are on (red). If they are, identify the defective elements. 3. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 4. Check that the RJ45 connectors connected between the battery modules are in good condition and tight. 5. Turn the batteries off and restart the equipment. 6. If the error persists, it will be because the BMS of the batteries is not working properly. This module must be replaced.
Alarm #26	Possible problem in the BMS of the batteries. 1. Click on the variable "VE. Reset" to reset the communication. For more information, refer to the manual. 2. If the alarm continues to occur, refer to the manual to perform a system shutdown (do not turn the batteries off). 3. Check that the batteries are working properly. Check to see if the error LEDs are on (red). If they are, identify the defective elements. 4. Check that the RJ45 connectors connected between the battery modules are in good condition and tightly tightened. 5. Check that the voltage between the DC terminals at the input of each of the inverters is sufficiently close to the voltage that appears on the battery display. 6. Turn the batteries off and restart the equipment. 7. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. To do this, you can observe the parameters of the battery and check that they are logical. 8. If the error persists, it will be because the BMS measurement of the batteries is not working properly. This module must be replaced.
Alarm #27	Possible problem in the BMS of the batteries. 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the batteries are working properly. Check to see if the error LEDs are on (red). If they are, identify the defective elements. 3. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 4. Check that the RJ45 connectors connected between the battery modules are in good condition and tightly tightened. 5. Turn the batteries off and restart the equipment. 6. If the error persists, it will be because the BMS of the batteries is not working properly. This module must be replaced.

INDICATOR	HELP
Alarm #28	Possible problem in the BMS of the batteries. 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the batteries are working properly. Check to see if the error LEDs are on (red). If they are, identify the defective elements. 3. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 4. Check that the RJ45 connectors connected between the battery modules are in good condition and tightly tightened. 5. Check the voltage between the DC bus from the EHR 5. Turn the batteries off and restart the equipment. 6. If the error persists, it will be because the BMS of the batteries or the contactor used for pre-loading is defective. Defective modules must be replaced.
Alarm #29	Possible problem in the BMS of the batteries. 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check the correct functioning of the batteries. Notice if the error LEDs (red) are running. If they are, identify the defective elements. 3. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 4. Check that the RJ45 connectors connected between the battery modules are in good condition and tightly tightened. 5. Check the voltage between the DC bus from the EHR 5. Turn the batteries off and restart the equipment. 6. If the error persists, it will be because the BMS of the batteries or the safety contactor is defective. Defective modules must be replaced.
Alarm #30	Possible problem in the BMS of the batteries. 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the batteries are working properly. Check to see if the error LEDs are on (red). If they are, identify the defective elements. 3. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 4. Check that the RJ45 connectors connected between the battery modules are in good condition. 5. Check the voltage between the DC bus from the EHR 5. Turn the batteries off and restart the equipment. 6. If the error persists, it will be because the BMS of the batteries or the contactor used for pre-loading is defective. Defective modules must be replaced.
Alarm #31	Possible problem in the BMS of the batteries. 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the batteries are working properly. Check to see if the error LEDs are on (red). If they are, identify the defective elements. 3. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 4. Check that the RJ45 connectors connected between the battery modules are in good condition and tightly tightened. 5. Turn the batteries off and restart the equipment. 6. If the error persists, it will be because the BMS of the batteries is not working properly. This module must be replaced.

INDICATOR	HELP
Alarm #32	Possible internal battery problem: 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 3. Check that the RJ45 connectors are in good condition and tight. 4. Check that the batteries are working properly. Check to see if the error LEDs are on (red). If yes, identify which battery is defective. 5. Turn the batteries off and restart the equipment. 6. If the error persists, it will be because one or more batteries are defective. This can be due both to deterioration of the cells and an internal communication failure.
Alarm #33	Possible problem in the BMS of the batteries. 1. Refer to the manual to perform a system repair (do not turn the batteries off). 2. Check that the batteries are working properly. Check to see if the error LEDs are on (red). If they are, identify the defective elements. 3. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 4. Check that the RJ45 connectors connected between the battery modules are in good condition and tightly tightened. 5. Check the voltage between the DC bus from the EHR 5. Turn the batteries off and restart the equipment. 6. If the error persists, it will be because the BMS of the batteries or the contactor used for pre-loading is defective. Defective modules must be replaced.
Alarm #34	Possible problem in the BMS of the batteries. 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the batteries are working properly. Check to see if the error LEDs are on (red). If they are, identify the defective elements. 3. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 4. Check that the RJ45 connectors connected between the battery modules are in good condition and tightly tightened. 5. Check the voltage between the DC bus from the EHR 5. Turn the batteries off and restart the equipment po. 6. If the error persists, it will be because the BMS of the batteries or the safety contactor is defective. Defective modules must be replaced.
Alarm #35	Possible internal battery problem: 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the RJ45 connectors are in good condition and tightly tightened. 3. Check that the batteries are working properly. Check to see if the error LEDs are on (red). If they are, identify which battery is defective. 5. Turn the batteries off and restart the equipment. 6. If the error persists, it will be because one or more batteries must be updated. Call technical support to perform the upgrade.

INDICATOR	HELP
Alarm #36	Possible internal battery problem: 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the RJ45 connectors are in good condition and tightly tightened. 3. Check that the batteries are working properly. Check to see if the error LEDs are on (red). If they are, identify which battery is defective. 5. Turn the batteries off and restart the equipment. 6. If the error persists, it will be because one or more batteries need attention. Call technical support to perform the upgrade.
Alarm #37	Possible problem in the BMS of the batteries. 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the batteries are working properly. Check to see if the error LEDs are on (red). If they are, identify the defective elements. 3. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 4. Check that the RJ45 connectors connected between the battery modules are in good condition and tight. 5. Check the voltage between the DC bus from the EHR 5. Turn the batteries off and restart the equipment. 6. If the error persists, it will be because the BMS of the batteries is defective. Defective modules must be replaced.
Alarm #38	There has been an excessive discharge of the battery, or the battery is defective. 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the batteries are working properly. Check to see if the error LEDs are on (red). If they are, identify the defective elements. 3. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 4. Check that the RJ45 connectors connected between the battery modules are in good condition and tightly tightened. 5. Turn the batteries off and restart the equipment. 6. Charge the batteries. You can charge batteries either by applying an operating mode or by performing a maintenance charge. 7. If the error persists, it will be because one or more cells of the batteries are defective. Defective modules must be replaced.
Alarm #39	There has been an excessive charge of the battery, or the battery is defective. 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the batteries are working properly. Check to see if the error LEDs are on (red). If they are, identify the defective elements. 3. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 4. Check that the RJ45 connectors connected between the battery modules are in good condition and tightly tightened. 5. Turn the batteries off and restart the equipment. 6. Operate the equipment in UPS mode and disconnect the input for approximately 15 minutes. 7. If the error persists, it will be because one or more cells of the batteries are defective. Defective modules must be replaced.

INDICATOR	HELP
Alarm #40	1. Check to see if the cooling is working. 2. If it is, disconnect the load at the output. 3. If the alarm persists, disconnect the primary power supply.
Alarm #41	1. Check that the heaters are working. 2. Wait until the system temperature rises to the desired level. 3. Once this temperature is reached, it can operate normally.
Alarm #42	You must exit the current operating mode and perform a maintenance charge until the battery is fully charged.
Alarm #43	Decrease the power consumption. If this is not possible, disconnect the load at the output.
Alarm #44	The temperature of the EHR is higher than recommended. Shut down the operating mode that is running and check that the fans are running. Wait 1 hour for the system temperature to drop.
Alarm #45	A battery balance error has occurred. 1. Refer to the manual to perform a system shutdown. (Do not turn off batteries) 2. Check that the batteries are working properly. Check to see if the error LEDs are on (red). 3. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 4. Turn the batteries off and restart the equipment. 5. If the error persists, it will be because one or more batteries are defective.
Alarm #46	Charge the battery. You can charge the batteries by applying an operating mode and performing a maintenance charge. If this is not possible, refer to the manual to see how to charge the batteries externally.
Alarm #47	A battery balance error has occurred. 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the batteries are working properly. Check to see if the error LEDs are on (red). If they are, identify the defective elements. 3. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 4. Check that the RJ45 connectors connected between the battery modules are in good condition and tightly tightened. 5. Turn the batteries off and restart the equipment. 6. If the error persists, it will be because one or more cells of the batteries are defective or the BMS cannot perform the balancing due to an error. Faulty modules need to be replaced
Alarm #48	There has been an excessive discharge of the battery, or the battery is defective. 1. Refer to the manual to perform a system shutdown (do not turn the batteries off). 2. Check that the batteries are working properly. Check to see if the error LEDs are on (red). If they are, identify the defective elements. 3. Check that the communication between the BMS of the batteries and the Cerbo GX is correct. 4. Check that the RJ45 connectors connected between the battery modules are in good condition and tightly tightened. 5. Turn the batteries off and restart the equipment. 6. Charge the batteries. You can charge the batteries by applying an operating mode and performing a maintenance charge. 7. If the error persists, it will be because one or more cells of the batteries are defective. Faulty modules need to be replaced

INDICATOR	HELP
Alarm #49	1. Check that the cooling is working. 2. If it is, disconnect the load at the output. 3. If the alarm persists, disconnect the primary power supply.
Alarm #50	1. Check that the heaters are active. 2. Wait until the system temperature rises to the desired level. 3. Once this temperature is reached, it can operate normally.
Alarm #51	1. The emergency mushroom has been pressed. 2. To restart the system, turn the mushroom until it is released.
Alarm #52	1. The differential switch has been activated. 2. Check that the EHR's input and output wiring is adequate and that all terminals are well threaded. 3. Check that the ground is connected as indicated in the manual and that the terminals are well threaded. 4. Once checked, reset the differential by clicking on "reset" to rest the magnetothermal output.
Alarm #53	The operating mode has shut down because the SoC of the battery is too low. Perform an optimal load or maintenance load. If this is not possible, refer to the manual to see how to perform an external load.
Alarm #54	The operating mode has shut down because the communication was not working properly. 1. Run the operating mode again. 2. If the error persists, check the RJ45 connectors inserted between the Cerbo GX and the CCLAN. 3. If the error persists, refer to the manual to perform a system shutdown (without disconnecting the batteries). 4. Restart the system. 5. If the error persists, it is due to a problem in the internal core, either the Cerbo GX or the eCORE.
Alarm #55	The system has shut down due to the fact that the maximum power contracted in the EHR output has been exceeded. Decrease the load in the installation and restart the desired functionality.

7.4 TROUBLESHOOTING

Question	Cause	Suggestion
The unit is turned on, but there is no output power.	The output contactor may not be turned on.	Check in the parameters tab that relay 2 is set to "CLOSE"
	The emergency stop may be activated.	Disable the emergency stop.
	The magnetothermal output may have tripped	Look at the current record and verify that it has tripped due to excess current
	Differential switch may have tripped	Check for any grounding and any current drifts that may have occurred and reset the differential switch
	The battery SoC is below 20% and no power source is connected to the EHR.	Connect the primary power supply and charge the battery; it is recommended that a maintenance charge be performed.
	The disconnectors between the inverter and the battery are turned off.	Set the disconnectors to "ON"
The external power supply is active (for example, the diesel generator is running) but the EHR is not synchronised.	The phase rotation of the diesel generator or the electrical distribution grid is not correct	Check the phase rotation of the generator or supply grid.
	Damaged cable between the diesel generator or the electrical distribution grid and EHR.	Check or replace cables and connectors.
	The magnetothermal output switch of the generator or the electrical distribution grid has tripped.	Verify that the magnetothermal output switch of the generator or electrical distribution grid is turned on.
	The input selector switch may be in the wrong position.	Check the position of the input selector switch behind the distribution door. Note that it has 3 positions: Off/Busbar and "Socket". Also make sure that the switch has been fully rotated to the right position.
The system power has been turned on, but it is not displayed on the HICORE	The HICORE fuse may have been damaged.	Replace the HICORE fuse.
	The 24 V DC regulator fuse may have been damaged.	Replace the 24 V DC regulator fuse.
Energy is being sent to the EHR, but no power is being charged or passed.	The AC magnetothermal input switch may have tripped	Check the position of the AC magnetothermal input switch and reset it if necessary.
The external power supply is active (for example, the diesel generator is running) but power only reaches the first phase (L1).	The maintenance load pin may be connected to the grid.	Verify that the maintenance plug is not connected
	Damaged cable between the diesel generator or the grid and EHR.	Test or replace cables and connectors.

Question	Cause	Suggestion
The output voltage is less than 400 V AC at a distance.	If the distance between the unit and consumers is more than 50 meters, the voltage may decrease.	Contact EHR Technical Services to arrange the adjustment of the system configuration.
0% SoC read and AC output DISCONNECTED.	This indicates that the storage voltage has reached its lower limit and has shut down its output to protect the batteries.	Charge the EHR from your chosen renewable energy source, from an AC grid connection, or from a secondary power system.
The output switch is activated continuously.	Connected consumers are consuming too much power for the unit's rated output.	Refer to the maximum output rating on the unit's nameplate and reduce the total power consumption accordingly.
The remote generator set start-up does not work.	The generator's remote start signal cable is loose or damaged.	Reconnect or replace the cable.
	Incorrect remote start-up switching mode (normally open/normally closed).	For more information about changing the switching mode, contact EHR Technical Services for assistance.
	The diesel generator set is in manual mode.	Switch the diesel generator set to automatic mode.
	Remote start-up of the generator set is not configured correctly.	Contact the technical services of the generator set manager.
	The remote control of the generator set may be set to manual.	Set the generator set remote control to automatic.
Diesel generator set running constantly.	The EHR may be in manual generator set mode.	Check the AC input fuses, connect them, or replace them if necessary.
	There can be a phase imbalance: too much energy is being consumed in one phase.	Consumers should be evenly distributed across the three stages as much as possible.
	The load is too high.	Reduce the load.
The RCD trips continuously.	There is an electrical failure with consumers connected to the AC output.	Check the electrical integrity of the consumers connected to the unit.
Output switch on the diesel assembly that is activated at start-up/shutdown.	There may be an RCD conflict.	Increase the activation current limit on the RCD of the diesel generator set and set the RCD variable to EHR. The adjustment of any RCD must be performed by a qualified electrician in accordance with applicable local rules and regulations.
The output voltage drops when the generator set shuts down.	The generator set output voltage is outside the acceptable voltage range for a smooth transition to inverter mode.	Check the phase voltage on the generator set to make sure it is in accordance with the unit's electrical standards (230 VAC or 120 VAC depending on the configuration).
In charging mode, the unit demands more power than is available in the generator set or power grid.	HICORE has the maximum power configured for the grid or generator set in the last working mode.	Please use the wizard to configure any working mode, and to set the maximum power of the generator set or the electrical grid to the current available power.

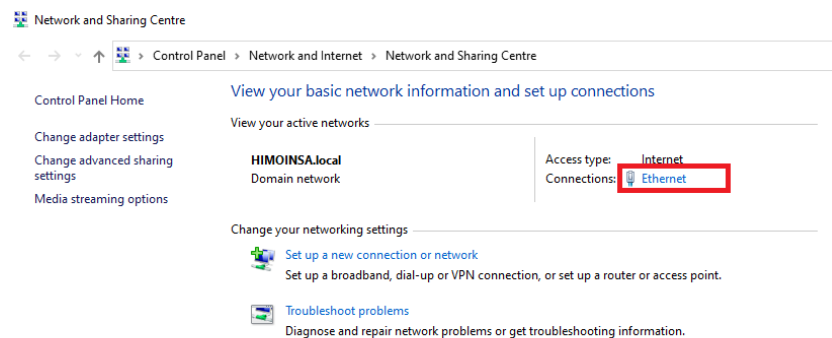
8. ANNEXES

8.1 CONFIGURING THE APN OF THE RUT350 ROUTER

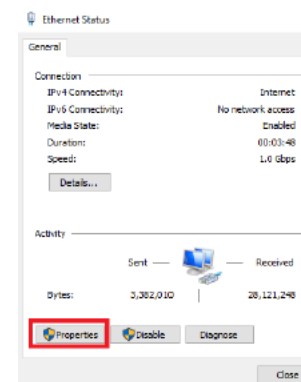
In the following instructions we will see the steps needed to configure the new SIM card of your local mobile operator.

1. STEP: Configure the DHCP:

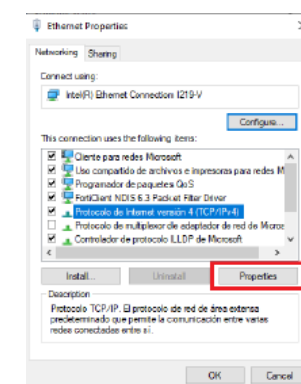
Connect the PC to any of the LAN1-LAN3 ports (not the WAN port) and configure the DHCP connection:



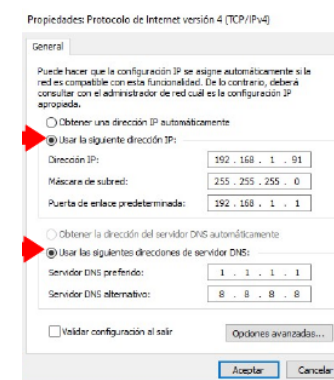
1



2



3



1. STEP: Access the router's web server

Once we have an ASSIGNED IP, start your internet browser and **open the page 192.168.1.1**



The image shows the Teltonika web interface for authorization. It features the Teltonika logo at the top, followed by the heading "Authorization Required". Below this, a message says "Please enter your username and password." There are two input fields: "Username" with the value "admin" and "Password" which is empty. A "Login" button is located below the password field.

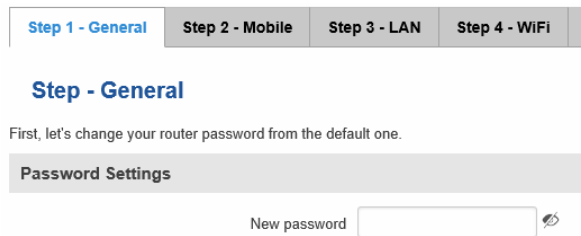
Usernameo: admin

Psasword: Himoinsa2022\$

2. STEP: Change Password

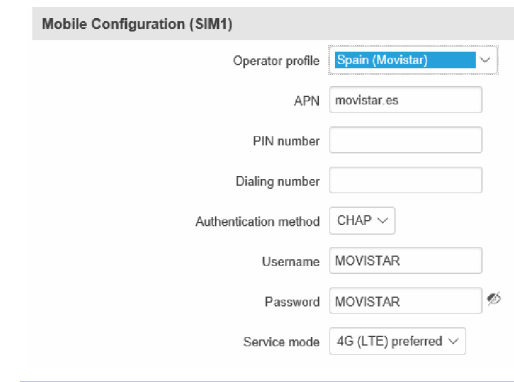
We can change the default password.

We will define our time zone (UTC), or we can synchronise with the browser.



This block contains two screenshots from a router's configuration interface. The top screenshot shows a tabbed menu with four options: "Step 1 - General" (selected), "Step 2 - Mobile", "Step 3 - LAN", and "Step 4 - WiFi". Below the tabs, the heading "Step - General" is followed by the instruction "First, let's change your router password from the default one." The bottom screenshot shows the "Password Settings" section with a "New password" input field and a toggle icon for password visibility.

3. STEP: Configuration SIM1



The image displays the "Mobile Configuration (SIM1)" settings page. It includes several configuration options: "Operator profile" set to "Spain (Movistar)", "APN" set to "movistar.es", "PIN number" (empty), "Dialing number" (empty), "Authentication method" set to "CHAP", "Username" set to "MOVISTAR", "Password" set to "MOVISTAR" with a toggle icon, and "Service mode" set to "4G (LTE) preferred".

1. Operator profile: we can select the operator or None to enter the fields manually.

2. APN: Configure the APN according to the mobile operator

3. PIN number: We recommend that you remove the PIN from the SIM to avoid 3 SIM and lock attempts (PUK unlock)

4. Dial number: leave empty.

5. Authentication method: Check with your mobile operator.

6. Username and password: check with your mobile operator.

7. Service mode: we can force 4G LTE, 3G, 2G connection As a preference, 4G LTE 3G or 2G, or we can select automatic mode.



FACTORIES

SPAIN • FRANCE • INDIA • CHINA • U.S.A. • BRASIL • ARGENTINA

SUBSIDIARIES

PORTUGAL | SINGAPORE | POLAND | UAE | PANAMA | GERMANY | ARGENTINA |
UNITED KINGDOM | DOMINICAN REPUBLIC | SOUTH AFRICA | MOROCCO | AUSTRALIA

HIPOWER SYSTEMS HEADQUARTERS

16600 South Theden Street

Olathe, Kansas, 66062

Main Tel.: (913) 495-5557 | Toll Free: (866) 710-2988

Fax: (913) 495-5557

www.hipowersystems.com

Disclaimers:

All information, illustrations and specifications in this manual are based on the latest information available at the time of publishing. The illustrations used in this manual are intended as representative reference views only. Moreover, because of our continuous product improvement policy, we may modify information, illustrations and/or specifications to explain and/or exemplify a product, service or maintenance improvement. We reserve the right to make any change at any time without notice.

All Rights Reserved:

No part of this publication may be reproduced or used in any form by any means - graphic, electronic, or mechanical, including photocopying, recording, taping, or information storage and retrieval systems - without the written permission of HIPOWER SYSTEMS.

Ver. 2 - 03/2024

HIPOWER®
A YANMAR COMPANY